

Name: _____

GCSE Chemistry Unit 2 Foundation Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2018					
2019					
2022					
2023					

Grade boundaries – Raw Score

Year	G	F	E	D	C	Max UMS
2018	17	21	25	30	35	40
2019	13	19	25	31	38	45
2022	19	23	27	31	36	41
2023	13	18	23	29	33	38

Surname	Centre Number	Candidate Number
Other Names		0

**GCSE – NEW**

3410U20-1



S18-3410U20-1

CHEMISTRY – Unit 2:
Chemical Bonding, Application of Chemical Reactions
and Organic Chemistry

FOUNDATION TIER

THURSDAY, 17 MAY 2018 – MORNING

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper
 you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen
 or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional
 page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **5** is a quality of extended response (QER) question where your writing skills will be
 assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common
 ions on the inside of the back cover.

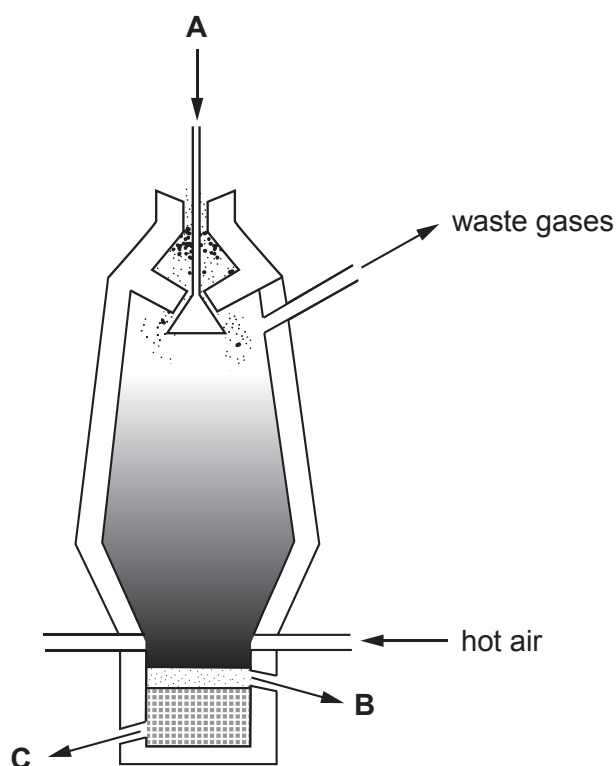
For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	12	
2.	10	
3.	10	
4.	9	
5.	6	
6.	13	
7.	8	
8.	12	
Total	80	

3410U201
01

MAY183410U20101

Answer all questions.

1. (a) The diagram shows where substances enter and leave the blast furnace in the extraction of iron from iron ore.



coke

slag

iron

limestone

iron ore

- (i) Use the substances in the box to complete the following sentences. [3]

The **three** raw materials which enter the furnace at **A** are ,
..... and

Product **B** is and product **C** is



(ii) One reaction that takes place in the furnace is



Underline the element which is removed from the iron(III) oxide during the reaction. [1]

iron

oxygen

carbon

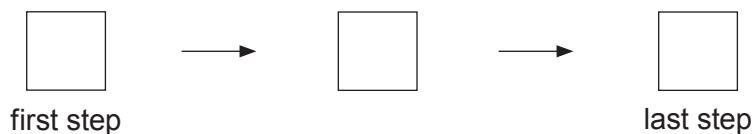
(b) Statements **D**, **E** and **F** show the three steps needed to prepare a sample of copper(II) chloride in the laboratory. The steps are **not** in the correct order.

D filter to remove excess copper(II) oxide

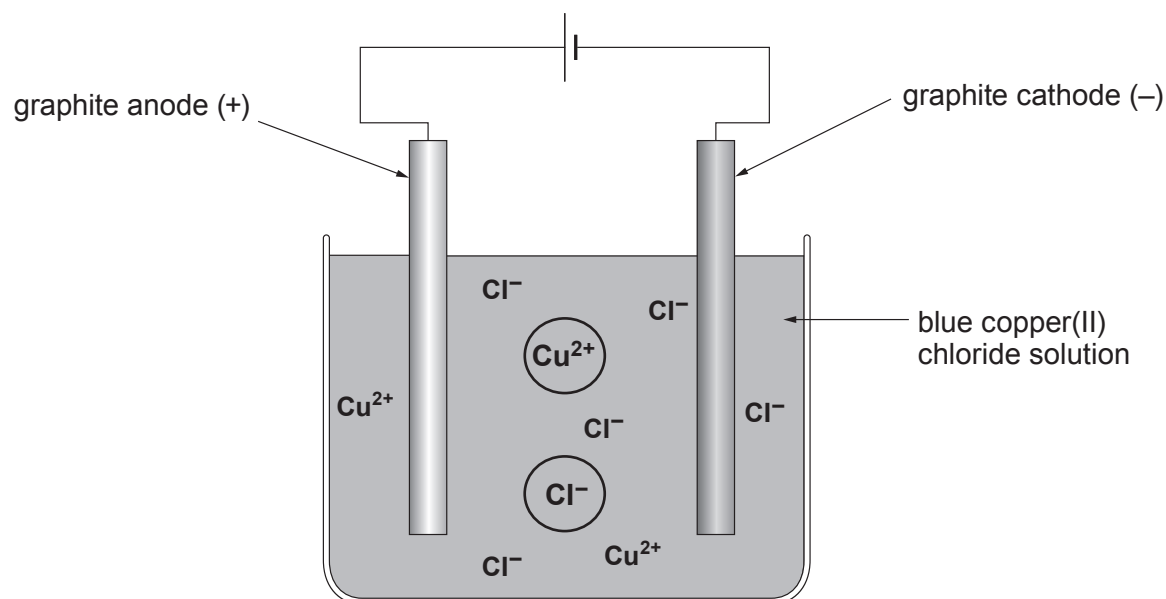
E leave the copper(II) chloride solution to evaporate at room temperature

F add excess copper(II) oxide to dilute hydrochloric acid

Complete the flow chart by putting the **letters** in the correct order. [2]



(c) The diagram shows the apparatus used to obtain copper from copper(II) chloride solution.



electrolyte	electrodes	electricity	electrolysis
-------------	------------	-------------	--------------

- (i) Choose words from the box to complete the following sentences. [2]

The breaking down of copper(II) chloride solution using an electric current is called

.....

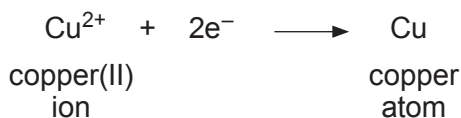
Electric current enters and leaves the copper(II) chloride solution by the

.....

- (ii) **By drawing arrows on the diagram**, show the direction in which the **circled** ions move during the process. [1]



(iii) The reaction occurring at the cathode is:



I. Put a tick (✓) in the box next to what e^{-} represents in the equation. [1]

electricity

☐

electron

☐

ion

☐

atom

☐

II. Copper(II) ions, Cu^{2+} , are removed from the solution during the process.

Put a tick (✓) next to the statement which describes what, if anything, happens to the colour of the copper(II) chloride solution. [1]

the blue solution turns darker

☐

the blue solution turns paler

☐

the blue solution turns yellow

☐

nothing happens to the colour of the blue solution

☐

(iv) Put a tick (✓) next to the gas which is formed at the anode. [1]

oxygen

☐

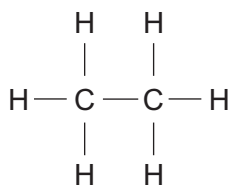
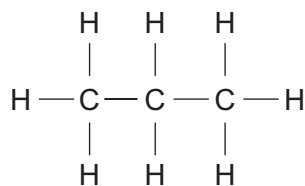
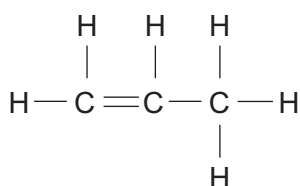
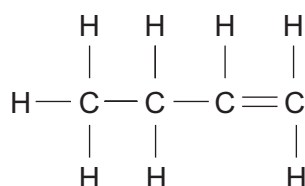
hydrogen

☐

chlorine

☐

2. (a) The structural formulae of four carbon compounds are shown below.

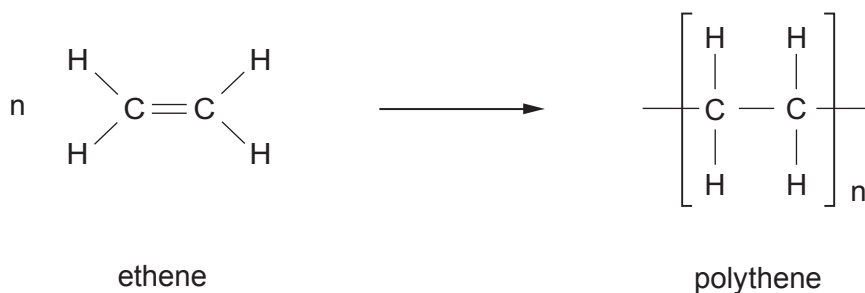
**A****B****C****D**

Complete the table by choosing the **letter A, B, C or D** which represents the structural formula of the named compounds. [2]

Name	Molecular formula	Structural formula
ethane	C_2H_6	
propene	C_3H_6	

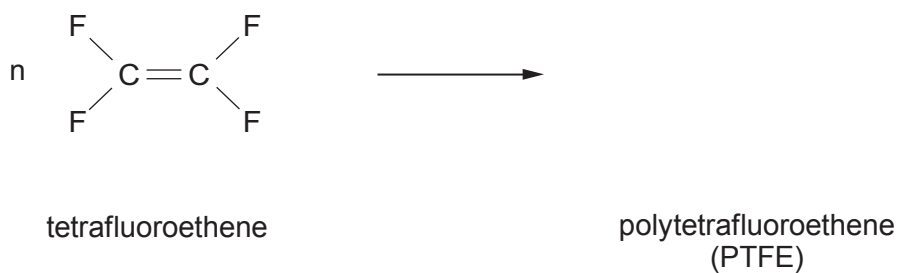


(b) The equation below shows the formation of polythene from ethene.

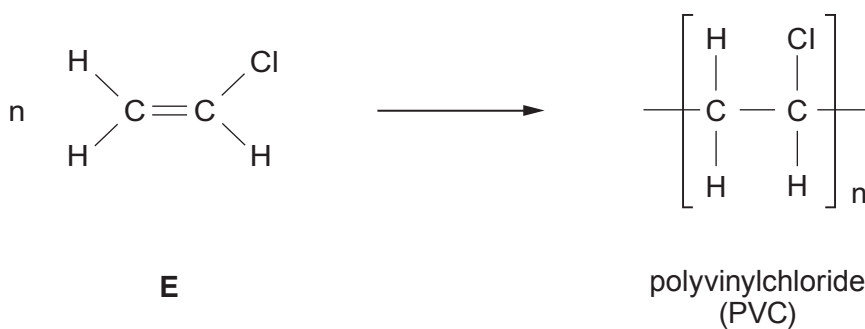


Use this information to complete parts (i) and (ii).

(i) Complete the equation for the formation of polytetrafluoroethene (PTFE) from tetrafluoroethene. [1]



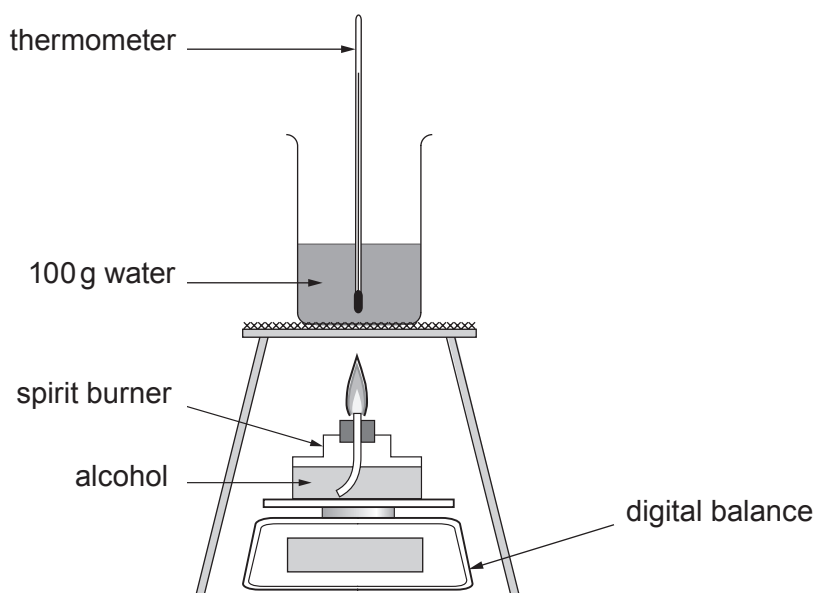
(ii) Name monomer **E**. [1]



E



- (c) A student carried out an investigation to find which alcohol gives out the most energy when burned.



100 g of water was heated by burning 1 g of each alcohol.

Alcohol	methanol	ethanol	propanol
Formula	CH ₃ OH	C ₂ H ₅ OH	C ₃ H ₇ OH
Relative molecular mass (M_r)	?	46	60
Temperature of water before heating (°C)	20	21	20
Temperature of water after heating (°C)	45	58	60
Energy given out (J)	?	15 700	17 200

- (i) Calculate the relative molecular mass (M_r) of methanol, CH₃OH.

[2]

$$A_r(\text{H}) = 1 \quad A_r(\text{C}) = 12 \quad A_r(\text{O}) = 16$$

$$M_r = \dots\dots\dots$$



- (ii) The energy given out can be calculated using the formula:

$$\text{energy given out} = \text{mass of water} \times 4.2 \times \text{temperature change}$$

Use the data given to calculate the energy given out when burning methanol. [2]

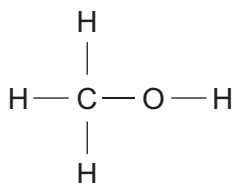
Energy given out = J

- (iii) Give the **letter** of the correct conclusion for the student's investigation. [1]

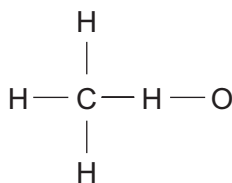
- A** all alcohols burn giving out the same amount of energy
- B** the greater the number of carbon atoms in the alcohol molecule the less energy is given out
- C** the greater the number of carbon atoms in the alcohol molecule the more energy is given out
- D** as the number of carbon atoms doubles the amount of energy given out doubles

Letter

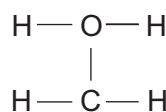
- (iv) Give the **letter** of the structural formula of methanol, CH₃OH. [1]



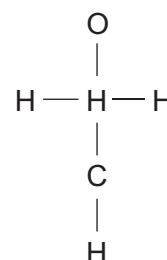
A



B



C



D

Letter



3. (a) Crude oil can be separated into simpler mixtures called fractions. These fractions contain hydrocarbon compounds called alkanes. **Table 1** shows information about some of the fractions obtained from crude oil by fractional distillation.

Fraction	Boiling point range (°C)	Number of carbon atoms present in the alkanes
petroleum gases	< 20	C ₁ -C ₄
petrol	30-75	C ₅ -C ₁₀
naphtha	70-170	C ₈ -C ₁₂
kerosene	170-250	C ₁₀ -C ₁₄
diesel oil	250-340	C ₁₄ -C ₂₄
lubricating oil	340-500	C ₂₁ -C ₃₀
fuel oil	490-580	C ₂₅ -C ₃₅
residue	> 580	>C ₃₅

Table 1

Use **only** the information in **Table 1** to answer parts (i)-(iii).

- (i) Hexane has a boiling point of 68 °C. Give the name of the fraction which contains hexane. [1]

.....

- (ii) One alkane is found in kerosene and in diesel. Give the number of carbon atoms in this alkane. [1]

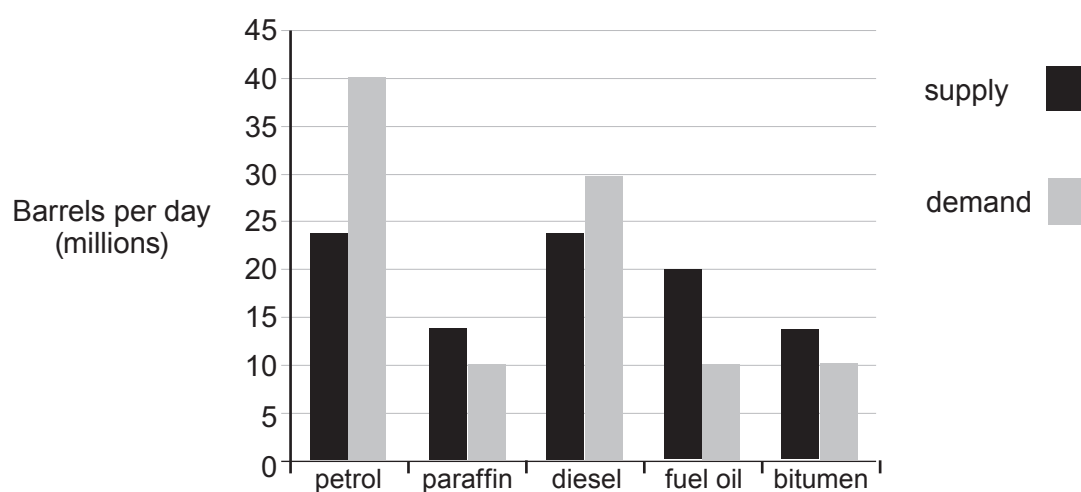
.....

- (iii) Give the number of carbon atoms in the alkane which has the **lowest** boiling point. [1]

.....



(b) The bar chart shows the **supply** and **demand** for some crude oil products.



Name the fractions for which the demand is greater than the supply. Suggest a reason why these fractions are in high demand. [2]

Fractions and

Reason

.....



- (c) Plastic carrier bags are made from polythene. Each plastic carrier bag can take 500-1000 years to decompose and may never break down in landfill. Paper bags are not necessarily an environmentally friendly alternative. Manufacturing paper bags wastes a lot of natural resources. Even starch-based biodegradable bags use natural resources during their manufacture.

Supermarkets give customers a choice of buying single-use or re-usable polythene carrier bags.

Table 2 shows the number of both types of plastic bag sold in UK supermarkets from 2011 to 2013.

Year	2011	2012	2013
	Number of bags (millions)		
Single-use bags	7977	8079	8455
Re-usable bags	415	408	445

Table 2

- (i) State **one** environmental problem related to the disposal of **all** types of carrier bag. [1]

.....

- (ii) Calculate the percentage of plastic bags sold in 2013 that were single-use bags. [2]

Percentage = %



- (iii) Although more plastic carrier bags were sold in 2013 than 2012, the total **mass** of those bags changed from 70 400 tonnes to 67 300 tonnes.

Put a tick (✓) in **two** boxes next to statements which could explain the reason for the change in mass. [2]

the bags were made the same thickness but from a less dense plastic

☐

customers re-used their plastic bags more often

☐

the bags were made from the same plastic but were thicker

☐

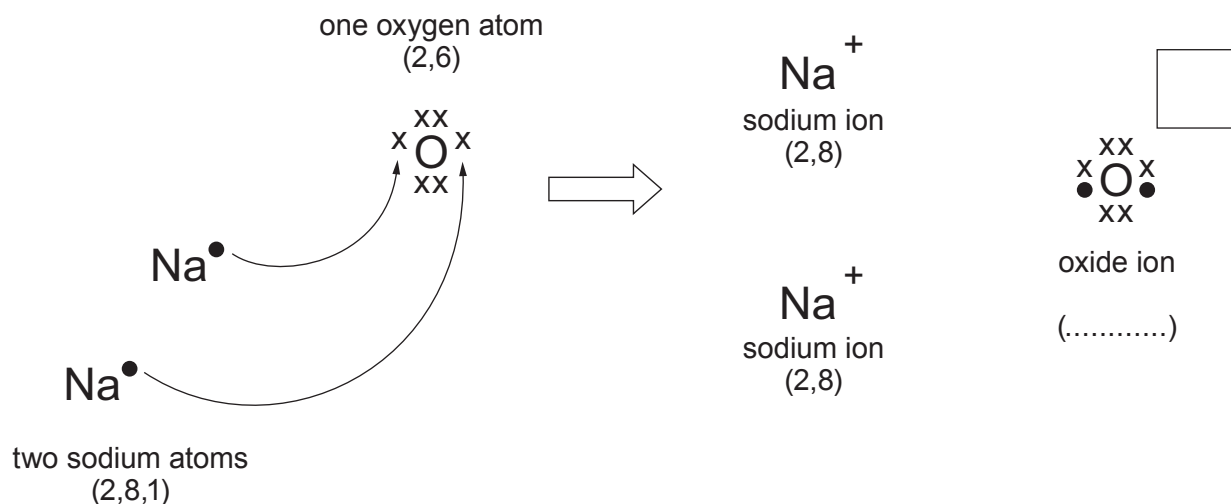
the bags were made from the same plastic but were thinner

☐

the bags were made the same thickness but from a more dense plastic

☐


4. (a) The diagram shows the electronic changes that occur when sodium reacts with oxygen to form sodium oxide. The ● and x symbols are outer shell electrons.



- (i) **Complete the diagram** by putting in the electronic structure **and** the charge of the oxide ion. [2]
- (ii) Complete and balance the **symbol** equation for the reaction between sodium and oxygen. [2]



- (b) The table shows the electronic structure of the elements present in water and hydrogen chloride.

Element	Electronic structure
hydrogen	1
oxygen	2,6
chlorine	2,8,7

The diagram shows the bonding in a water molecule.



Give the **letter** of the diagram which shows the bonding in a hydrogen chloride molecule. [1]



A



B



C

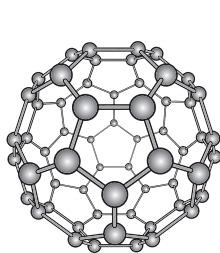


D

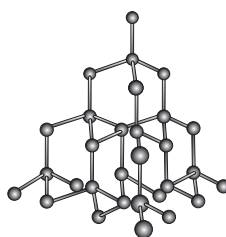
Letter



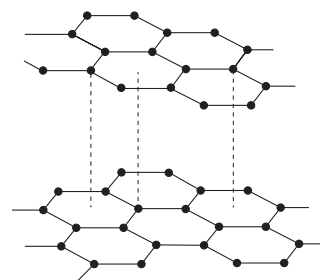
(c) Fullerene, diamond and graphite are different forms of carbon.



fullerene



diamond



graphite

(i) Underline the structure of these three forms of carbon.

[1]

simple molecular

giant covalent

giant ionic

(ii) The box shows some properties of different forms of carbon.

transparent

insoluble in water

conducts electricity

hollow

soft

does not conduct electricity

hard

Complete the table by choosing the property which makes the different forms of carbon suitable for the uses shown. [3]

Form of carbon	Use	Property
diamond	drill bits	
graphite	pencils	
fullerene	drug delivery	



Examiner
only

5. Every year thousands of acres of moorland are destroyed by fires in Wales. Firefighters use several methods to put out this type of fire.



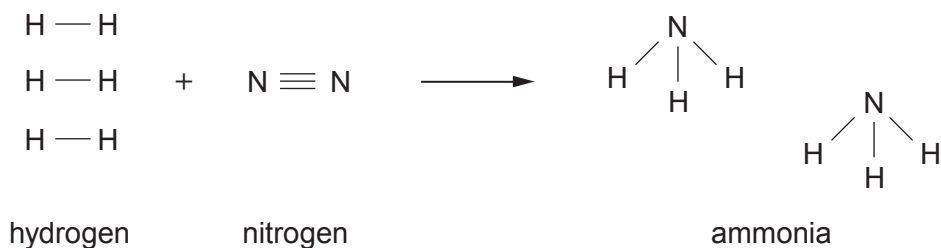
State and explain, in terms of the fire triangle, **three** methods that are used to put out moorland fires. Each method must refer to a **different** part of the fire triangle. [6 QER]

6



6. Ammonia is manufactured from hydrogen and nitrogen in the Haber Process.

- (a) The equation shows the bonds which are broken and the bonds which are formed during the manufacture of ammonia.



The relevant bond energies are shown in the table.

Bond	Bond energy (kJ)
$\text{N} \equiv \text{N}$	945
$\text{H} - \text{H}$	436
$\text{N} - \text{H}$	391

- (i) Calculate the **total** energy needed to break **all** the bonds in the hydrogen molecules and the nitrogen molecule. [2]

Energy = kJ

- (ii) Calculate the **total** energy released when **all** the bonds in the ammonia molecules are formed. [2]

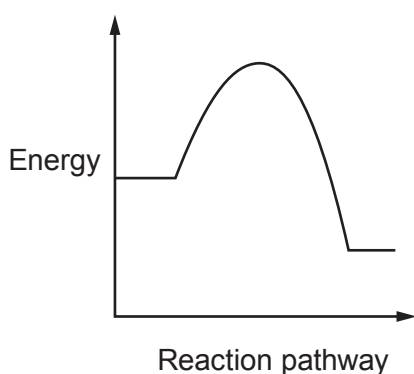
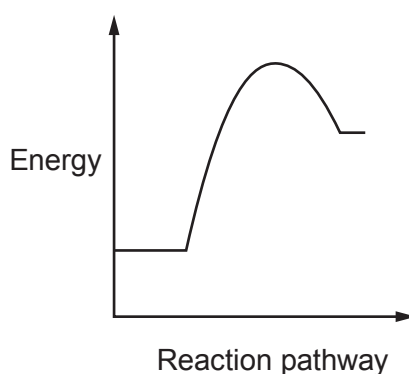
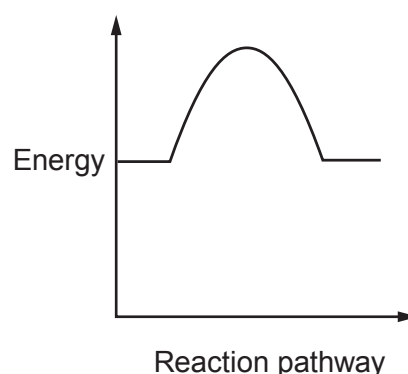
Energy = kJ

- (iii) Use your answers to parts (i) and (ii) to calculate the overall energy change. [1]

Energy change = kJ



- (b) The manufacture of ammonia is an exothermic reaction. Put a tick (✓) in the box next to the energy profile diagram that shows an exothermic reaction. [1]


☐

☐

☐

- (c) The table shows the percentage yield of ammonia under different pressure and temperature conditions.

Pressure (atm)	Temperature (°C)				
	100	200	300	400	500
	Yield of ammonia (%)				
100	96.7	81.7	52.5	25.2	10.6
200	98.4	89.0	66.7	40.0	18.3
400	99.4	94.6	79.7	55.4	31.9

Use the information in the table to answer parts (i) and (ii).

- (i) State what happens to the yield of ammonia as the temperature increases. [1]

- (ii) One manufacturer carries out the Haber Process at 200 atm and 450 °C.

Underline the approximate percentage yield of ammonia formed under these conditions. [1]

10 %

30 %

40 %

58 %



(d) Ammonia is used in the manufacture of nitrogenous fertilisers. One example of a nitrogenous fertiliser is ammonium nitrate. Ammonium nitrate is formed by the reaction between an acid and ammonia.

(i) Complete the **word** equation by naming the acid used in this reaction. [1]

ammonia + $\longrightarrow$ ammonium nitrate

(ii) When ammonium sulfate solution is warmed with sodium hydroxide solution a pungent gas is formed. Damp red litmus paper is used to test this gas.

I. Describe the change in colour of the litmus paper during the test. [1]

.....

II. State the property of this gas which causes the colour change. [1]

.....

III. Name the gas formed. [1]

.....

(iii) Nitrogenous fertilisers pollute streams and rivers. State how nitrogenous fertilisers get into these waterways. [1]

.....

.....



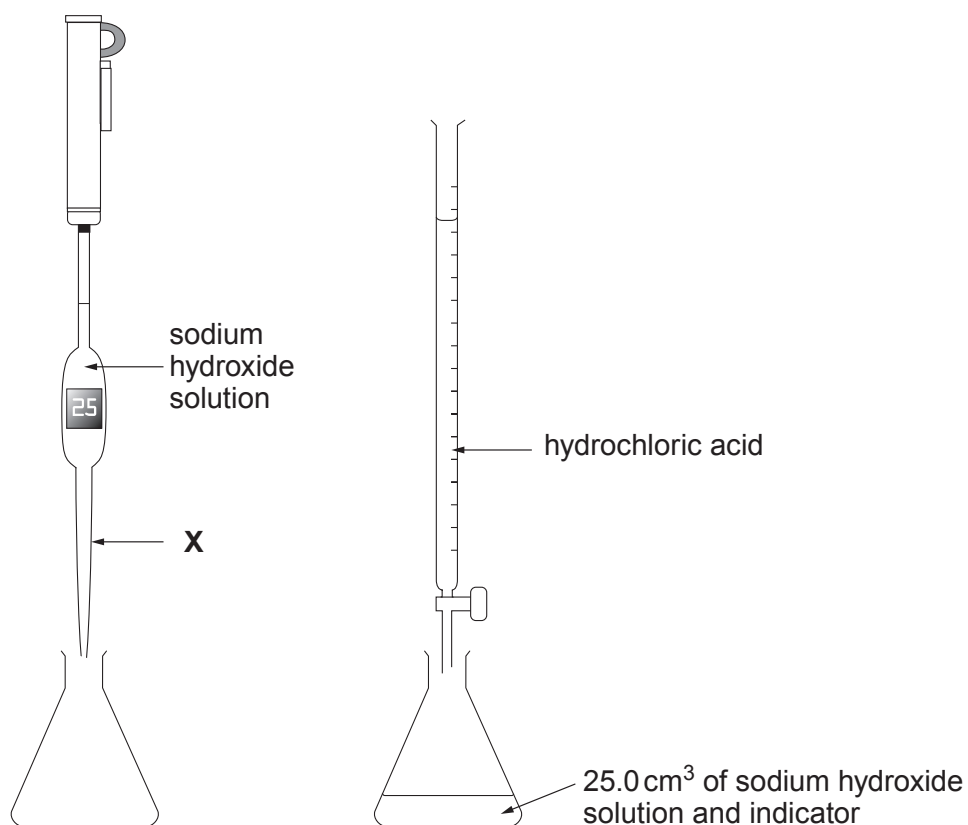
7. (a) Hydrochloric acid, HCl , reacts with sodium hydroxide solution to form sodium chloride and water only.

Write a balanced **symbol** equation for this reaction.

[2]

- (b) A group of students was asked to find the volume of hydrochloric acid solution needed to neutralise 25.0 cm^3 of sodium hydroxide solution. They decided to titrate sodium hydroxide with hydrochloric acid.

Apparatus



Results

Titre	1	2	3
Volume of hydrochloric acid needed (cm^3)	18.2	17.8	18.0



- (i) Name the piece of apparatus **X**. [1]

.....

- (ii) Explain the purpose of the indicator. [1]

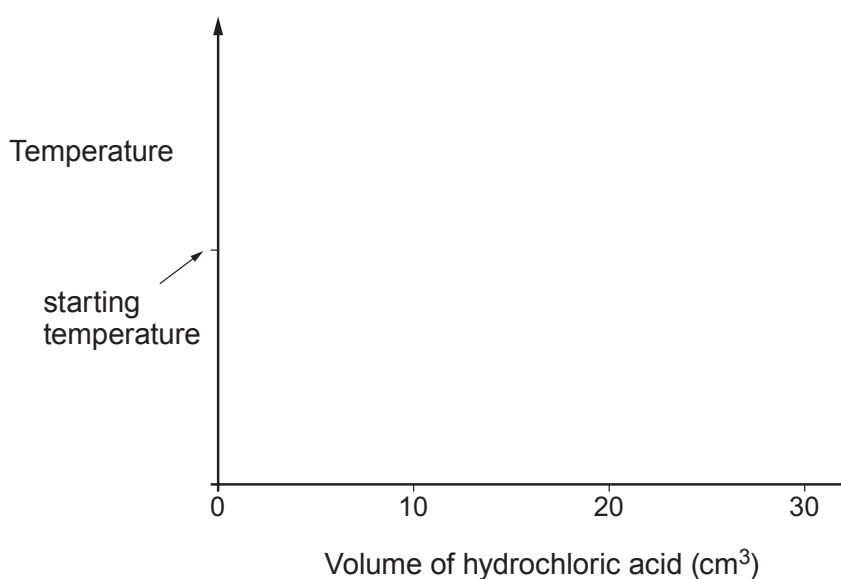
.....

- (iii) Calculate the mean volume of hydrochloric acid needed to neutralise 25.0 cm^3 of the sodium hydroxide solution. [1]

Mean volume = cm^3

- (iv) The change in temperature during the reaction can be monitored using a temperature sensor.

Sketch a graph on the axes below to show how the temperature changes as more and more acid up to a total of 30 cm^3 is added. [2]



- (v) The experiment was repeated using hydrochloric acid of **half** the original concentration.

State the volume of hydrochloric acid that would be needed to change the indicator colour. [1]

Volume = cm^3



8. (a) When a mixture of iron(III) oxide and aluminium powder (Thermit mixture) is heated, there is a violent reaction. The reaction is carried out in a tube surrounded by a mound of sand because the temperature reaches 2500 °C. A bead of iron is recovered from the sand. The picture below shows the reaction taking place in a darkened room.



- (i) Give the reason why the iron formed in the reaction is molten. [1]

.....
.....

- (ii) Complete and balance the **symbol** equation for this reaction. [2]



- (iii) State which of the substances is oxidised. Give the reason for your choice. [1]

.....
.....

- (iv) When a mixture of magnesium oxide and aluminium powder is heated, there is no reaction.

List iron, magnesium and aluminium in order of reactivity. [1]

Most reactive

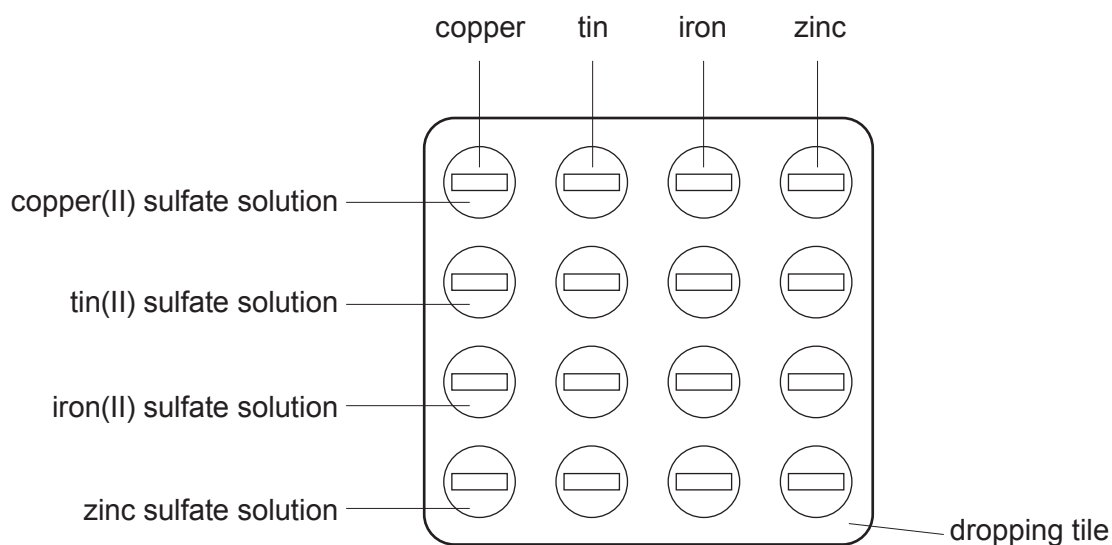
.....

Least reactive



- (b) Some metals are more reactive than others. A more reactive metal displaces a less reactive metal from its compounds.

A student was given tin, iron, copper and zinc and solutions of the metal sulfates. Using a dropping pipette, she put a little of one of the sulfate solutions in four of the depressions of the dropping tile. She did this for each solution in turn. She then put a piece of metal foil in each of the solutions, as shown below.



- (i) Put a tick (✓) next to the question which **best** describes the investigation the student is carrying out. [1]

Which displacement is the most exothermic?

☐

Which metal can displace copper from solution?

☐

What is meant by the reactivity series?

☐

What are the positions of the four metals in the reactivity series?

☐

- (ii) The student recorded the results by putting a tick (✓) next to a mixture which showed signs of a reaction and a cross (X) next to a mixture which showed no signs of a reaction.

The student concluded that:

tin displaces copper
iron displaces tin
iron displaces copper
zinc displaces iron

Give the **letter** of the tile below which shows the results she recorded.

[1]

Letter

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

A

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

B

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

C

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

D

- (iii) Another student said that not all of the tests were necessary. Give **one** example of a test not needed. Explain your choice.

[2]

Example

Explanation

.....



(c) Copper displaces silver from a solution of silver nitrate, AgNO_3 , to form copper(II) nitrate solution.

(i) Describe **one** change the student would **see** during this displacement reaction. [1]

.....

.....

(ii) Write a balanced **symbol** equation for this reaction. [2]

.....

12

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		



2

Group

3

1 H Hydrogen 1																			
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10		
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18		
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36		
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54		
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86		
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89																	
																		Key	

Key

relative atomic mass

A_r	Symbol	Name	Z
1	H	Hydrogen	1
2	He	Helium	2
3	Li	Lithium	3
4	Be	Beryllium	4
5	B	Boron	5
6	C	Carbon	6
7	N	Nitrogen	7
8	O	Oxygen	8
9	F	Fluorine	9
10	Ne	Neon	10
11	Na	Sodium	11
12	Mg	Magnesium	12
13	Al	Aluminum	13
14	Si	Silicon	14
15	P	Phosphorus	15
16	S	Sulfur	16
17	Cl	Chlorine	17
18	Ar	Argon	18
19	K	Potassium	19
20	Ca	Calcium	20
21	Sc	Scandium	21
22	Ti	Titanium	22
23	V	Vanadium	23
24	Cr	Chromium	24
25	Mn	Manganese	25
26	Fe	Iron	26
27	Co	Cobalt	27
28	Ni	Nickel	28
29	Cu	Copper	29
30	Zn	Zinc	30
31	Ga	Gallium	31
32	Ge	Germanium	32
33	As	Arsenic	33
34	Se	Selenium	34
35	Br	Bromine	35
36	Kr	Krypton	36
37	Rb	Rubidium	37
38	Sr	Strontium	38
39	Y	Yttrium	39
40	Zr	Zirconium	40
41	Nb	Niobium	41
42	Mo	Molybdenum	42
43	Tc	Technetium	43
44	Ru	Ruthenium	44
45	Rh	Rhodium	45
46	Pd	Palladium	46
47	Ag	Silver	47
48	Cd	Cadmium	48
49	In	Indium	49
50	Sn	Tin	50
51	Pb	Lead	82

atomic number

Surname	Centre Number	Candidate Number
Other Names		0

**GCSE**

3410U20-1



S19-3410U20-1

THURSDAY, 16 MAY 2019 – MORNING**CHEMISTRY – Unit 2:****Chemical Bonding, Application of Chemical Reactions and Organic Chemistry****FOUNDATION TIER**

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **6(a)** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	4	
2.	9	
3.	8	
4.	13	
5.	7	
6.	11	
7.	8	
8.	9	
9.	7	
10.	4	
Total	80	

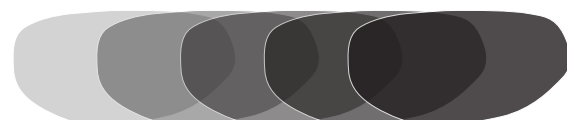
3410U201
01

MAY193410U20101

Answer all questions.

1. (a) The diagrams show some uses of smart materials which rely on their unusual properties.

sunglasses



cloudy

sunny

baby bath thermometer



cold



hot

car bumpers



nappies



The box contains the names of four types of smart material.

thermochromic pigment

shape memory polymer

hydrogel

photochromic pigment

Use the information given to complete the table. [2]

Unusual property	Use	Type of smart material
it regains its original shape on heating	car bumpers	shape memory polymer
it changes colour with changing temperature		
it can absorb up to 1000 times its volume of water		

3410U201
03



(b) Draw a line to link the use of each nano-material to its property.

[2]

Examiner
only

Use

Property

nano-silver in dressings for
cuts and burns

antibacterial

blocks harmful UV light

nano-titanium dioxide in
sunscreen creams

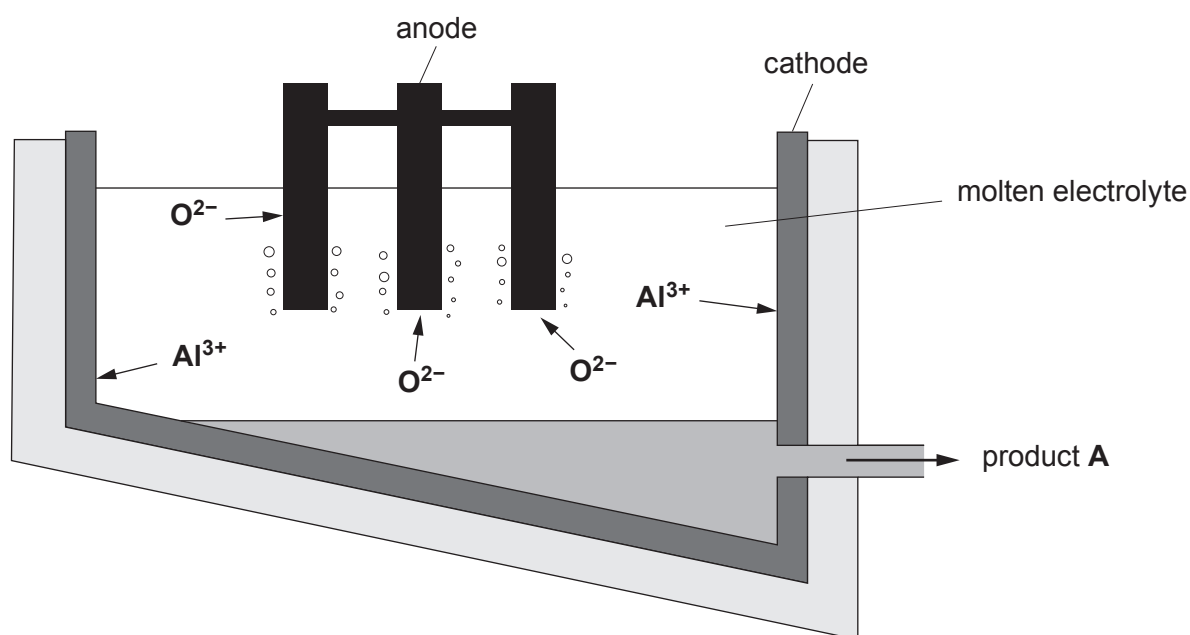
breaks down dirt

strong and light

4



2. The diagram shows an electrolysis cell used in the extraction of aluminium from alumina.



bauxite	electrical	molten alumina	positive
negative	cryolite	molten aluminium	oxygen
			light

(a) Choose words from the box to complete the following sentences. [5]

The molten electrolyte is a mixture of alumina and

Product **A** is

Oxygen is formed at the anode which is the electrode.

Alumina is obtained from an ore called

The process of extracting aluminium is expensive because it uses a lot of energy.

(b) Balance the symbol equation that shows the overall reaction. [1]

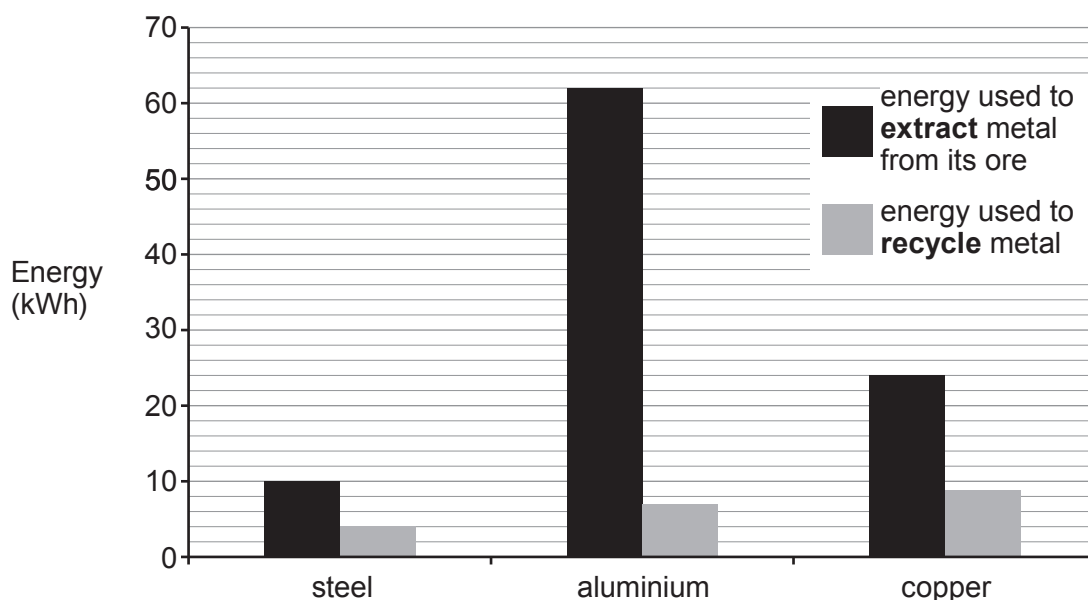


- (c) 250 tonnes of an aluminium ore can produce 195 tonnes of alumina.
Calculate the percentage of alumina in the ore.

[2]

Percentage = %

- (d) The chart compares the amount of energy used to extract and to recycle three metals.



Use the information in the chart to answer the following question.

Put a tick (✓) in the box next to the statement that describes why it is more cost effective to recycle aluminium than steel and copper. [1]

The energy used to extract metals is greater than that used in recycling them

☐

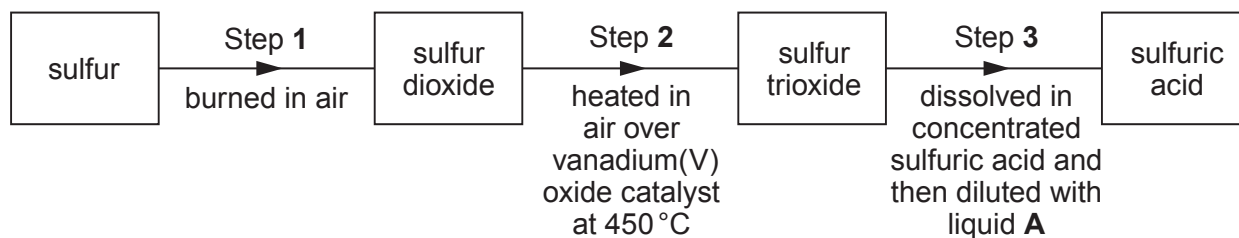
The difference between the energy used to extract and the energy used to recycle is the greatest

☐

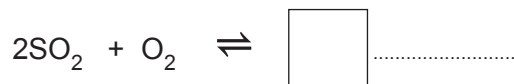
The energy used in recycling is less than for copper but greater than for steel

☐

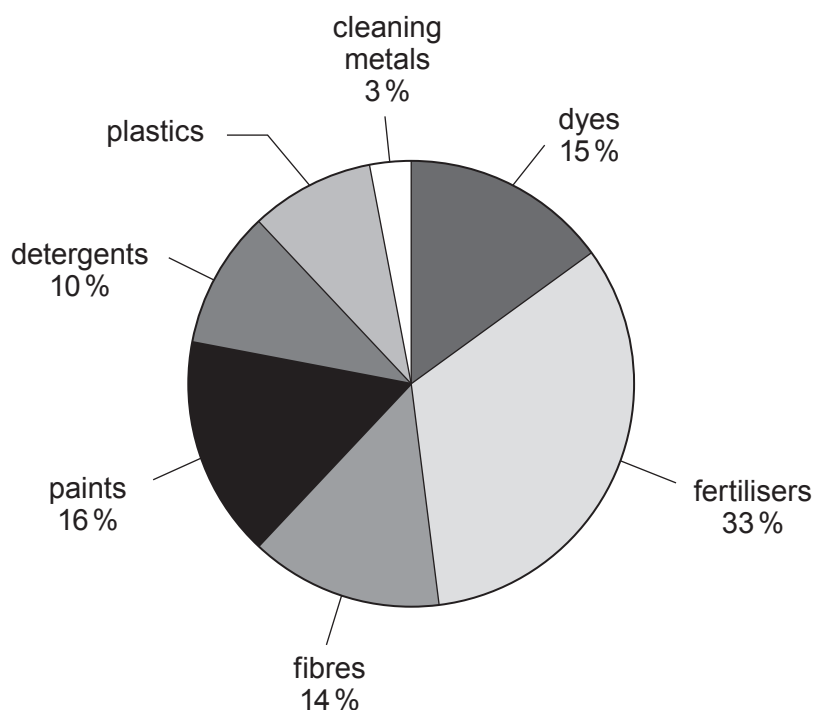

3. (a) The flow diagram shows steps in the manufacture of sulfuric acid using the Contact Process.



- (i) Name the **element** in the air which reacts with sulfur in step 1. [1]
.....
- (ii) State the purpose of the catalyst in step 2. [1]
.....
- (iii) Name liquid **A** used to dilute the concentrated sulfuric acid in step 3. [1]
.....
- (iv) Complete and balance the symbol equation for the reaction in step 2. [2]



(b) The pie chart shows the main uses of sulfuric acid.



Calculate the percentage of sulfuric acid used for making plastics.

[2]

Percentage = %

(c) One important use of sulfuric acid is in the production of the fertiliser ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$.

Complete the following word equation for the production of ammonium sulfate.

[1]

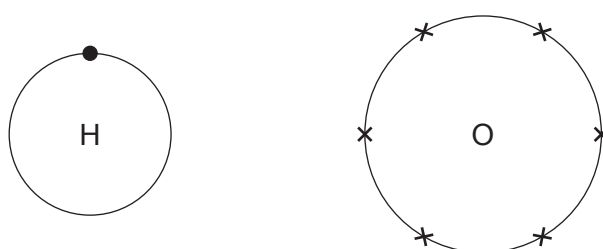
sulfuric acid + $\longrightarrow$ ammonium sulfate



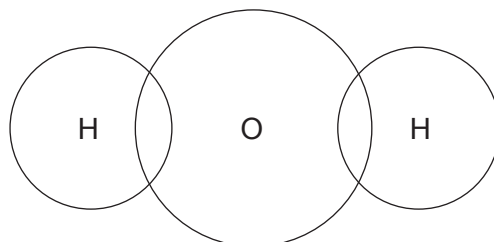
4. (a) The table shows the electronic structure of the elements present in water.

Element	Electronic structure
hydrogen	1
oxygen	2,6

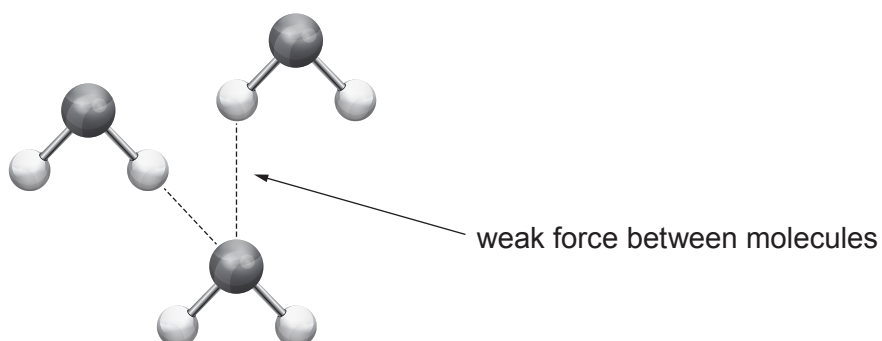
The diagrams show the outer shell electrons in an atom of hydrogen and an atom of oxygen.



- (i) Complete the diagram to show the outer shell electrons in a molecule of water. [2]



(ii) The diagram shows some water molecules and the weak forces between them.

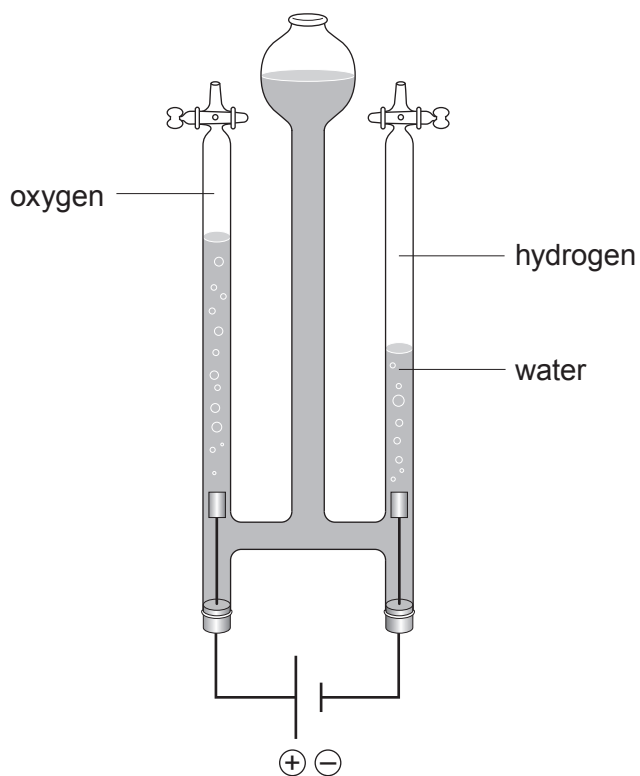


Put a tick (✓) in the box next to the property of water which can be explained by the weak forces between water molecules. [1]

poor conductor of electricity	
colourless	
good conductor of heat	
low melting point and boiling point	



- (b) The diagram shows the apparatus used by a group of students to investigate the volume of hydrogen and oxygen gas formed during the electrolysis of water.



The overall equation for the electrolysis of water is as follows.



- (i) The table shows the total volume of hydrogen formed over 10 minutes.

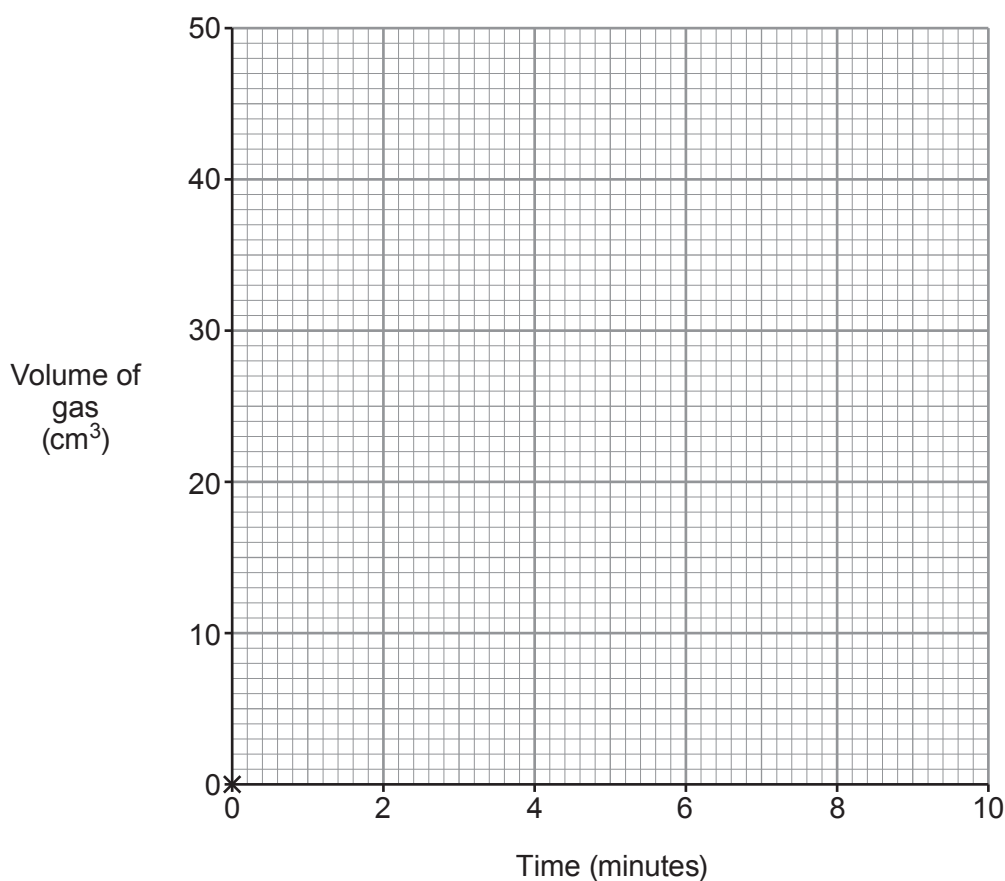
Time (minutes)	0	2	4	6	8	10
Volume of hydrogen (cm ³)	0	10	20	30	40	50



- I. Plot the values from the table on the grid. Draw a suitable line. Label this line '**hydrogen**'.

(0,0) has been plotted for you.

[2]



- II. Using a ruler, draw a second line **on the grid** to show the volume of oxygen that would be collected during the same 10 minutes. Label this line '**oxygen**'.

[1]

- III. Describe the relationship between the volume of hydrogen and the volume of oxygen formed during electrolysis.

[2]

.....

.....

.....

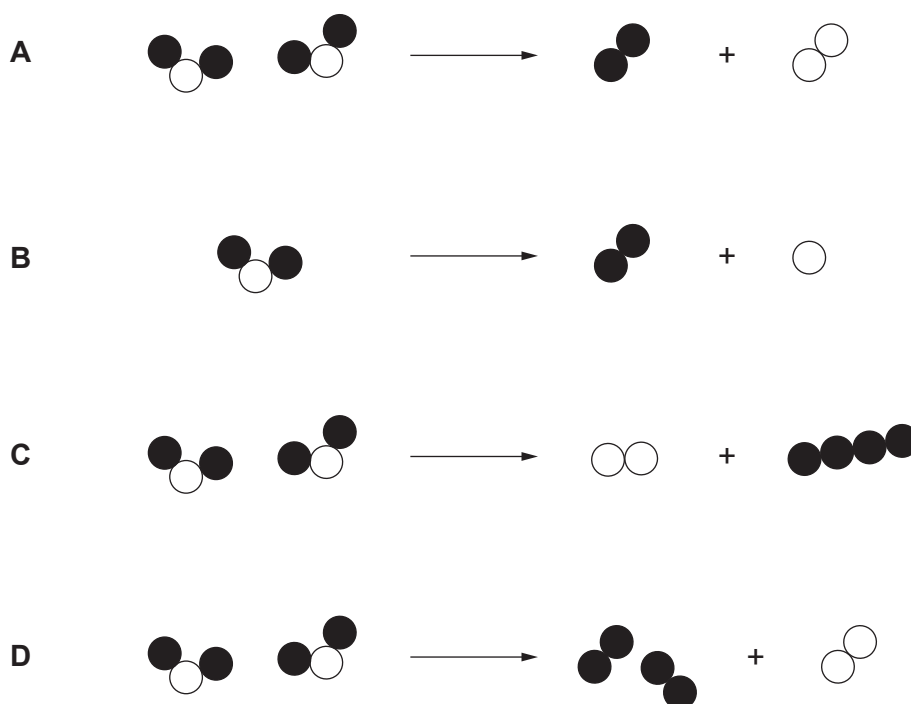


(ii) Electrolysis of water can be represented by the following equation.



Give the **letter, A, B, C or D**, of the diagram which also represents this reaction.

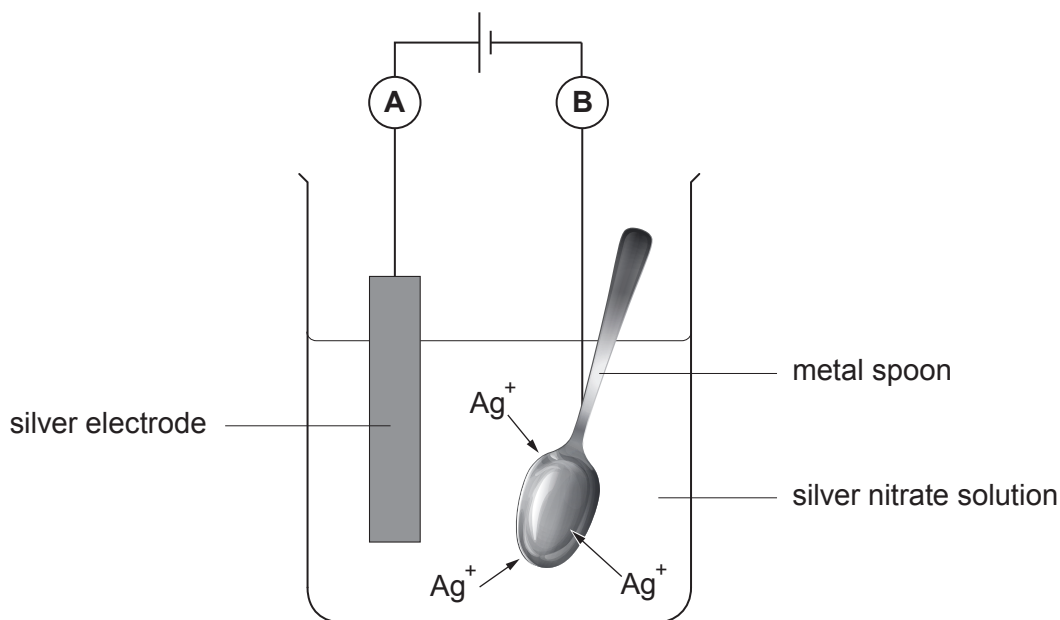
[1]



Letter



- (c) One use of electrolysis is in electroplating. The diagram shows the silver plating of a metal spoon.



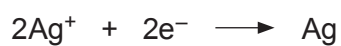
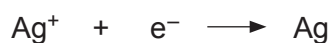
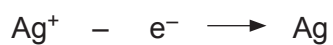
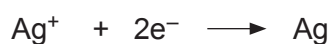
- (i) State why this process does not work for a plastic spoon. [1]

.....

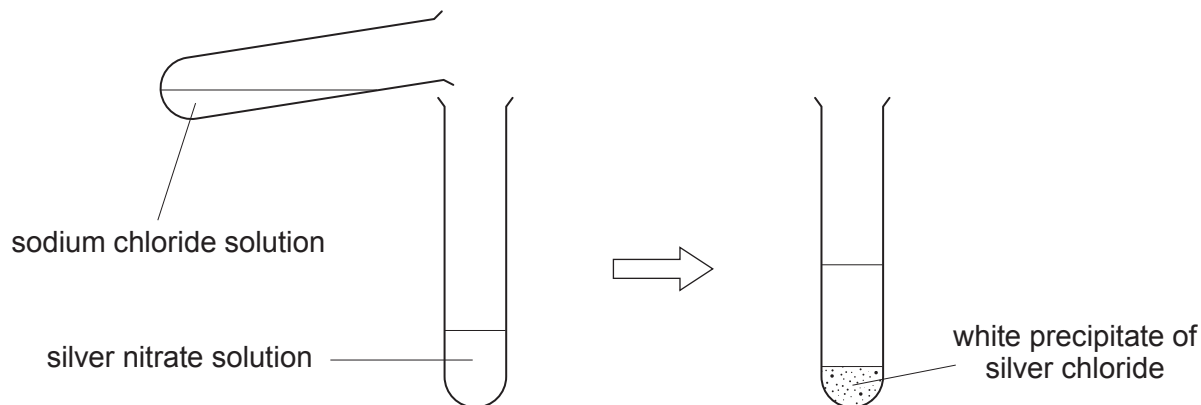
- (ii) Explain why the silver ions move towards the spoon. [2]

.....

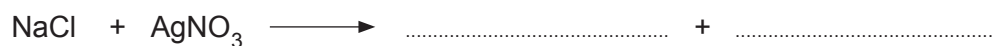
- (iii) Put a tick (✓) in the box next to the electrode equation for the reaction at electrode **B**. [1]


☐

☐

☐

☐


5. (a) Dafydd was asked to make some silver chloride. He formed a white precipitate of silver chloride by mixing solutions of sodium chloride and silver nitrate.



- (i) Complete the symbol equation for this reaction. [1]



- (ii) Put a tick (✓) in the box next to the statement which describes why this method works. [1]

silver is more dense than sodium

☐

silver chloride is soluble

☐

silver chloride is insoluble

☐

silver is below sodium in the reactivity series

☐

- (iii) Give the name of the process that you would use to separate the precipitate of silver chloride from the reaction mixture. [1]

.....



- (b) Calculate the relative formula mass (M_r) of silver nitrate, AgNO_3 . [2]

$$A_r(\text{O}) = 16$$

$$A_r(\text{N}) = 14$$

$$A_r(\text{Ag}) = 108$$

$$M_r = \dots\dots\dots$$

- (c) The relative formula mass (M_r) of sodium chloride, NaCl , is 58.5.

Calculate the percentage of sodium in sodium chloride. Give your answer to **1 decimal place**. [2]

$$A_r(\text{Na}) = 23$$

$$\text{Percentage} = \dots\dots\dots \%$$



6. (a) Plastics have replaced many traditional materials. The diagrams show plastics being used for three different types of food and drink packaging and containers.

Fruit packaging



Drinks containers



Take-away meal containers



Describe the advantages and disadvantages of using plastics in these examples.

[6 QER]

.....

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(b)

Microplastics in the Ocean

Plastics are used in many areas of modern life.

Figure 1 shows the amount of plastic production between 1950 and 2013.

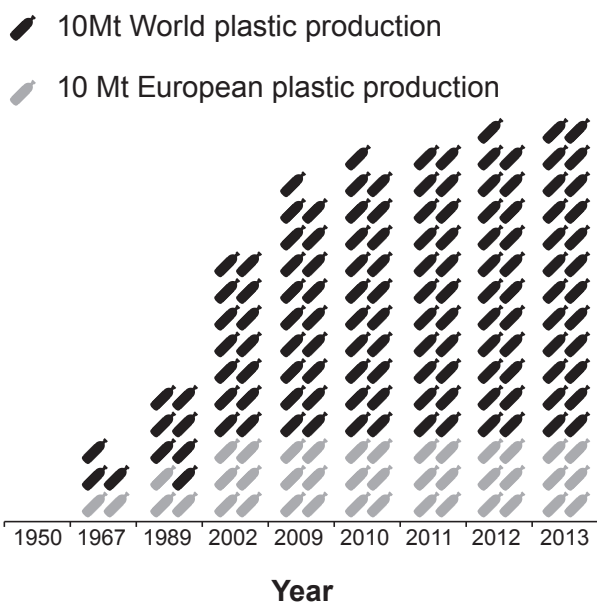
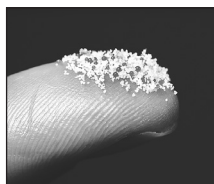


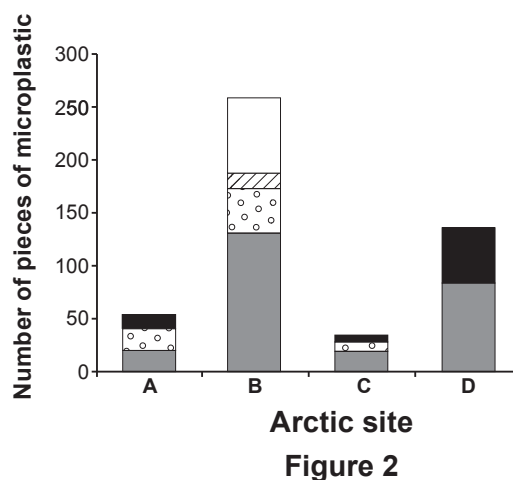
Figure 1

One of plastic's greatest properties, its durability, is also one of the main reasons that plastics present a threat to the marine environment.

The term 'microplastic' is used to describe plastic particles that are less than 5 millimetres in diameter, which includes particles as small as 10 nanometres. Microplastics can be found in some cosmetic products, toothpastes and soaps.



Microplastics are spread throughout the oceans and are found on shorelines from the Arctic to Antarctica. **Figure 2** shows the number of microplastic pieces found in sea ice at four Arctic sites during a survey in 2014.



Key Plastic

-  nylon
-  polyester
-  rayon
-  polypropylene
-  others

Microplastics have been found inside the bodies of marine animals. Microplastics often contain chemicals that can absorb poisons such as pesticides from the surrounding seawater. There is strong evidence of transfer of poisons from eaten microplastics into animal tissues. Nano-size microplastics have been shown to cross cell membranes, under laboratory conditions, causing tissue damage.

Public awareness of the potential for microplastics to damage marine animals is low compared with that of the impact of plastic litter in our seas and oceans. Effective education of society is essential to raise awareness of the damaging effects of microplastics.



- (i) Put a tick (✓) in the box next to the size of microplastics. [1]

less than 10 mm

☐

between 5 mm and 10 nm

☐

greater than 5 mm and less than 10 nm

☐

between 5 mm and 10 mm

☐

- (ii) Put a tick (✓) in the box next to the statement that best describes the amount of plastic produced in Europe since 2002. [1]

plastic production has remained constant

☐

plastic production has increased

☐

plastic production has decreased

☐

- (iii) Name the type of plastic most often found in the Arctic microplastic survey. [1]

.....

- (iv) Put a tick (✓) in the box next to a hypothesis which needs further testing by scientists. [1]

the quantity of microplastics found in the Earth's oceans is increasing

☐

microplastics carry contaminants from sea water into animals

☐

microplastics cause tissue damage in marine animals

☐

microplastics are a greater problem near land than in deep water

☐

- (v) Suggest a method of educating people of the hazards of microplastics. [1]

.....

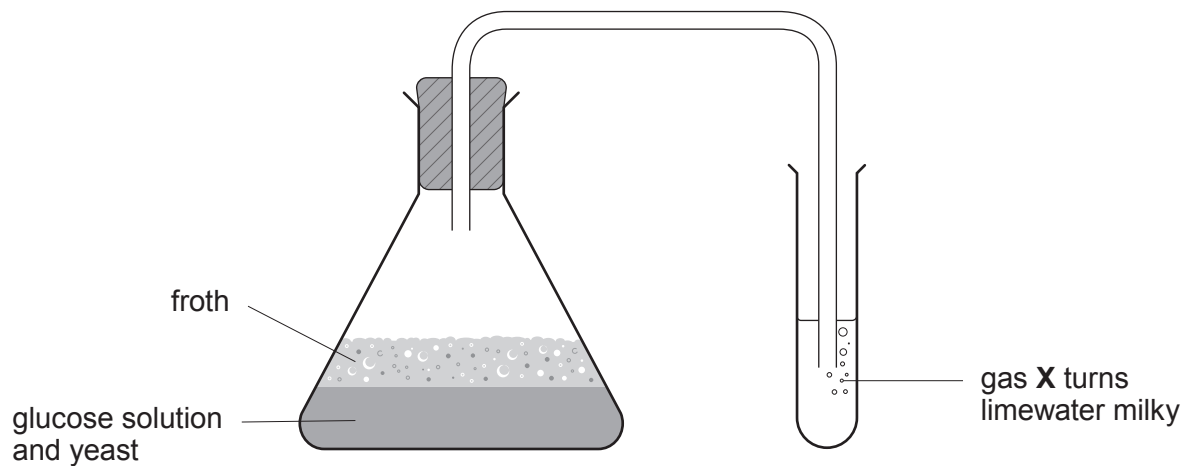


7. (a) The table shows the names, molecular formulae and structural formulae of some alcohols. Complete the table. [3]

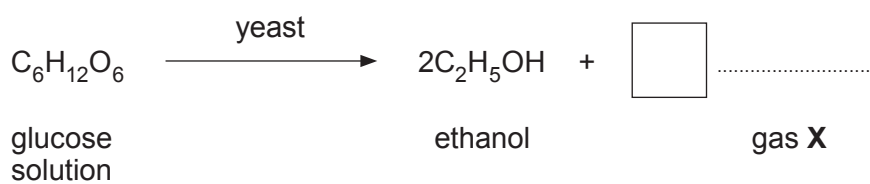
Name	Molecular formula	Structural formula
methanol		$ \begin{array}{c} \text{H} \\ \\ \text{H} - \text{C} - \text{O} - \text{H} \\ \\ \text{H} \end{array} $
ethanol	$\text{C}_2\text{H}_5\text{OH}$	
.....	$\text{C}_3\text{H}_7\text{OH}$	$ \begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{H} & \\ & & & & & & \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - \text{O} - \text{H} \\ & & & & & & \\ & \text{H} & & \text{H} & & \text{H} & \end{array} $



(b) The diagram shows apparatus that can be used to make ethanol by fermentation.



(i) Complete and balance the symbol equation for the reaction. [2]



(ii) Yeast acts as a catalyst in the process. Give the reason why catalysts are written above the arrow in equations. [1]

.....



Examiner
only

- (c) In Brazil sugar cane is used to make ethanol which can be used instead of petrol in cars. Many people see ethanol as the fuel of the future but others are concerned with environmental and social issues.

The table shows information relating to the burning of 1 dm³ of ethanol and petrol.

	Ethanol	Petrol
Source	sugar cane	crude oil
Energy released (MJ)	23.5	33.0
CO ₂ released (kg)	1.5	2.2

Use the information in the table and your knowledge to answer the following question.
Explain **one** advantage and **one** disadvantage of using ethanol instead of petrol in cars. [2]

Advantage

.....

.....

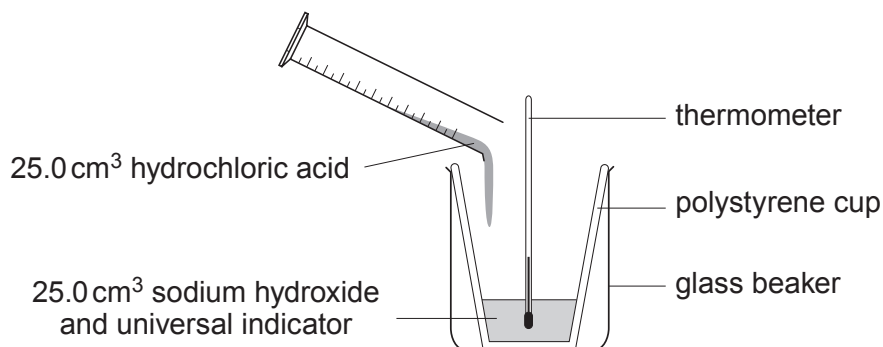
Disadvantage

.....

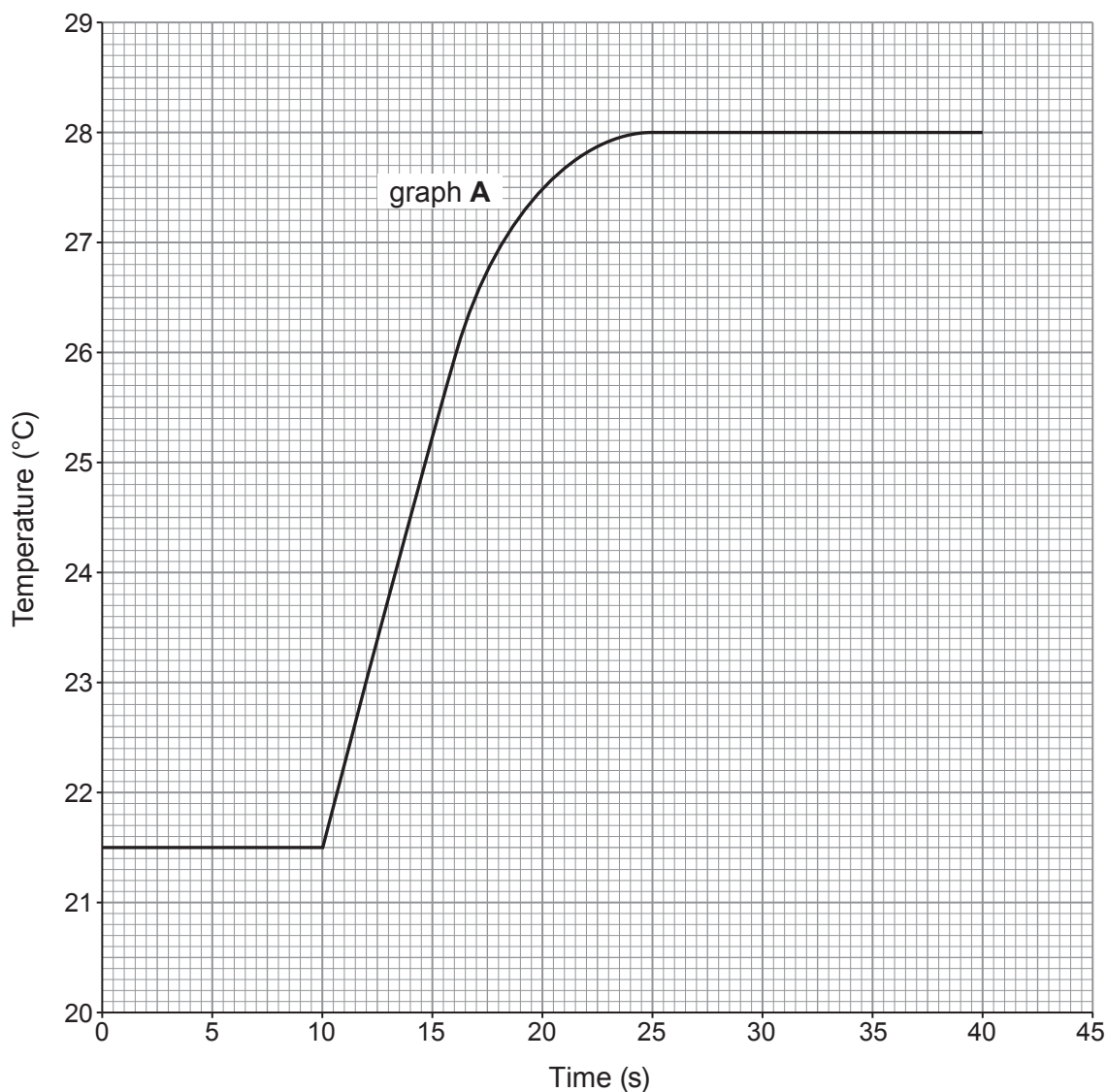
.....



8. A student investigated the temperature rise during a neutralisation reaction.



The student put 25.0 cm³ of sodium hydroxide solution and 5 drops of universal indicator into a polystyrene cup and recorded the temperature of the alkali. After 10 seconds the student added 25.0 cm³ of dilute hydrochloric acid to the alkali and recorded the temperature every 5 seconds for another 30 seconds. Graph **A** shows the results obtained.



- (a) (i) Use the graph to find the maximum temperature rise during the reaction. [1]

Temperature rise = °C

- (ii) The energy given out can be calculated using the following formula.

energy given out = **total** volume of reaction mixture $\times$ 4.2 $\times$ temperature rise

Calculate the energy given out during the reaction. [2]

Energy given out = J

- (iii) The temperature of the contents in the cup was recorded after 2 hours. Give the final temperature reading you would expect. Give the reason for your answer. [1]

Final temperature °C

Reason

.....

- (b) The student repeated the experiment using 25.0 cm³ of ethanoic acid of the same concentration as the hydrochloric acid. The table shows the results obtained.

Time (s)	0	5	10	15	20	25	30	35	40
Temperature (°C)	21.5	21.5	21.5	24.0	26.0	26.9	27.0	27.0	27.0

Plot the results on the grid on page 26. Draw a suitable line. Label your graph **B**. [2]

- (c) Use the graphs to state which of the two acids is the stronger – hydrochloric acid or ethanoic acid. Give the reason for your choice. [1]

Acid

Reason

.....



(d) The temperature rises in both experiments were much **lower** than expected. The student suggested that using a temperature sensor instead of a thermometer would give temperature rises closer to the expected values.

(i) State why using a temperature sensor would still give a lower than expected temperature rise. [1]

.....

.....

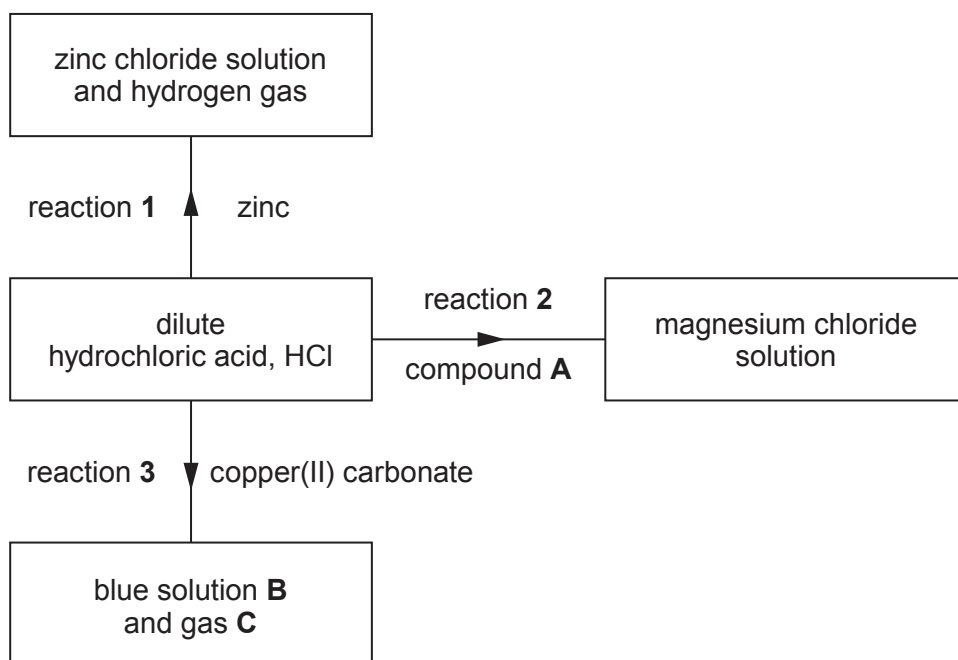
(ii) What improvement to the apparatus would you suggest to the student to obtain temperature rises closer to the expected values? [1]

.....

.....



9. The diagram shows three reactions which are used to prepare soluble salts.



- (a) (i) Name compound **A**. [1]

.....

- (ii) Give the names of blue solution **B** and gas **C** formed in reaction 3. [2]

blue solution **B**

gas **C**

- (b) Write the balanced symbol equation for reaction 1. [2]

.....

- (c) Reaction 1 was repeated using magnesium instead of zinc. Explain the difference, if any, that you would expect to see. [2]

.....

.....

.....



10. The tables show the molecular formulae of some alkanes and alkenes.

Alkanes	Alkenes
CH_4	C_2H_4
C_2H_6	C_3H_6
C_3H_8	
C_4H_{10}	

- (a) The general formula for the alkene family is C_nH_{2n} . Give the general formula for the alkane family. [1]

.....

- (b) When alkanes and alkenes completely burn in air they form the same two products. Give the chemical formulae for both products. [1]

..... and

- (c) Draw the structural formula for propene. [1]

- (d) Bromine water is used to distinguish alkenes from alkanes. Describe the colour **change** seen when bromine water is added to an alkene. [1]

.....

4

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1

2

3

4

5

6

7

0

1 H Hydrogen 1																				4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10			
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18			
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36			
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54			
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86			
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89																		

Key

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE – CONTINGENCY**

3410U40-1



Z22-3410U40-1

THURSDAY, 23 JUNE 2022 – MORNING

CHEMISTRY – Unit 2:
Chemical Bonding, Application of Chemical Reactions
and Organic Chemistry

FOUNDATION TIER

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
 You may use pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **5(a)** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	5	
2.	10	
3.	9	
4.	8	
5.	10	
6.	9	
7.	9	
8.	9	
9.	11	
Total	80	

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Answer **all** questions.

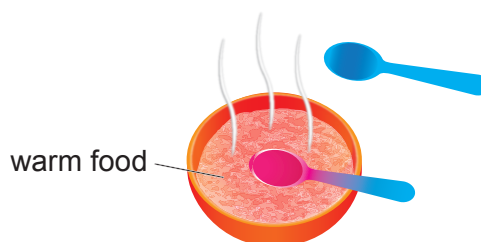
1. Smart materials are widely used in everyday life.

(a) The diagrams show some uses of smart materials which rely on their unusual properties.

nappies



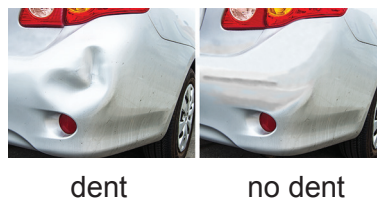
baby-feeding spoon



summer T-shirts



car bumper



The box contains the names of four different types of smart material.

thermochromic pigment

shape memory polymer

hydrogel

photochromic pigment



Use the information opposite to complete the table below.

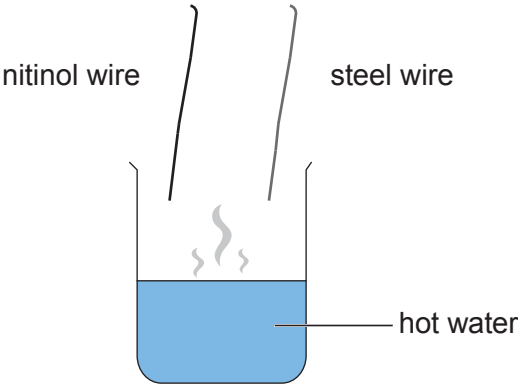
[4]

Unusual property	Use	Type of smart material
Able to absorb many times its own volume of water	nappies	hydrogel
Changes colour with changing temperature		
Changes colour with changing light intensity		

(b) Paperclips can be made from nitinol and from steel. Nitinol is a shape memory alloy but steel is not.



A student uncoiled both paperclips into straight wires and dropped them into hot water.



Put a **tick** (✓) in the box next to the statement that describes what happens.

[1]

- both wires reform into the shape of a paperclip

only the nitinol wire reforms into the shape of a paperclip

only the steel wire reforms into the shape of a paperclip

neither wire reforms into the shape of a paperclip
- ☐

☐

☐

☐



2. (a) The diagram shows the colours of universal indicator at different pH values.

Colour	red	orange	yellow	green	blue	navy blue	purple
pH	1 - 2	3 - 4	5 - 6	7	8 - 9	10 - 11	12 - 13

- (i) Sodium hydroxide solution turns universal indicator purple.

Put a **tick** (✓) in the box next to the correct statement.

[1]

sodium hydroxide is a strong alkali

☐

sodium hydroxide is a weak alkali

☐

sodium hydroxide is a strong acid

☐

sodium hydroxide is a weak acid

☐

- (ii) Ethanol is neutral. State the **colour** universal indicator would turn in ethanol.

[1]

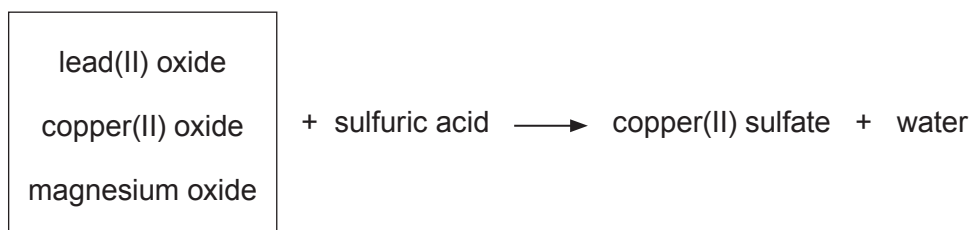
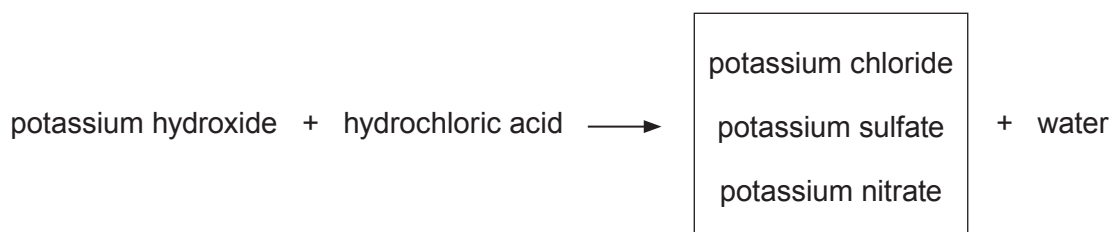
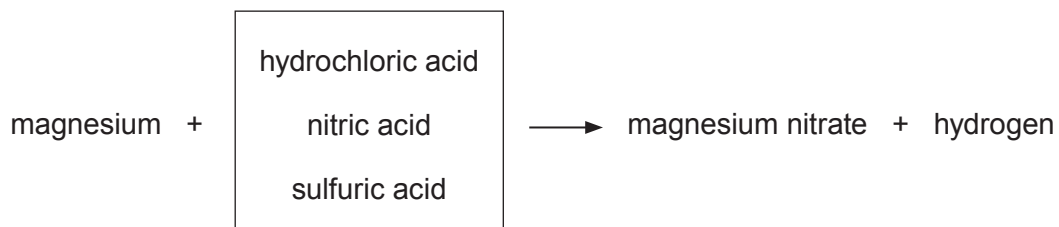
.....



(b) The word equations show reactions of acids with a metal, an alkali and a base.

Circle the correct substance in the box to complete each equation.

[3]



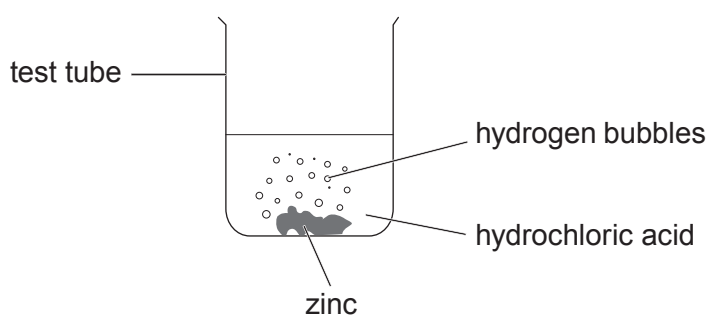
- (c) A student was asked how he would prepare a sample of zinc chloride crystals from zinc. The word equation for the reaction is



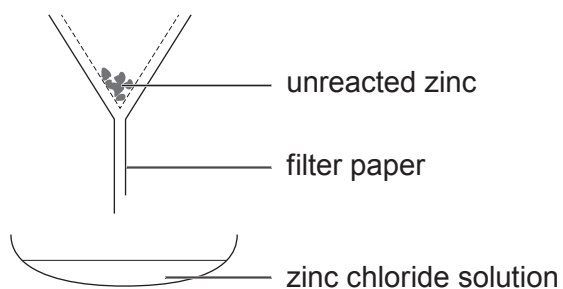
The diagrams below were taken from the student's work book.

There is **one** error in **each** of the student's diagrams.

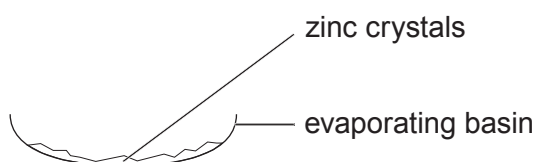
Step 1



Step 2



Step 3



(i) **Circle** the error in each diagram. [3]

(ii) Zinc chloride contains the ions Zn^{2+} and Cl^- .

Underline the correct formula of zinc chloride. [1]



(iii) Put a **tick** (✓) in the box next to the test to show that the gas given off is hydrogen. [1]

the gas pops with a lighted splint

☐

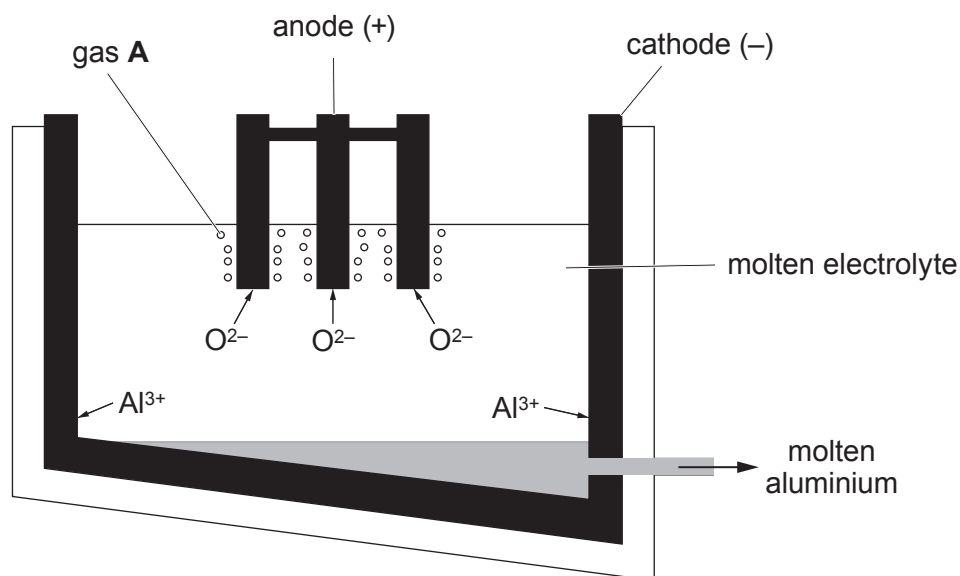
the gas turns limewater milky

☐

the gas relights a glowing splint

☐

3. (a) The diagram shows a cell used in the extraction of aluminium from aluminium oxide.



positive	oxygen	cryolite	negative
aluminium	hydrogen	steel	graphite

- (i) Choose words from the box to complete each of the following sentences. [4]

The electrolyte is a mixture of molten aluminium oxide and

Aluminium ions, Al^{3+} , are attracted to the electrode with a charge.

Gas A is

The electrodes are made of

- (ii) Balance the equation for the overall reaction. [1]



- (b) Calculate the relative formula mass (M_r) of aluminium oxide, Al_2O_3 . [2]

$$A_r(\text{O}) = 16$$

$$A_r(\text{Al}) = 27$$

Relative formula mass =

- (c) The box below contains some properties of aluminium.

low density	non-magnetic	resists corrosion
good thermal conductor	good electrical conductor	

Window frames can be made from different materials.

Choose **two** properties from the box which make aluminium a **better** material than iron for making window frames. [2]

Property 1

Property 2



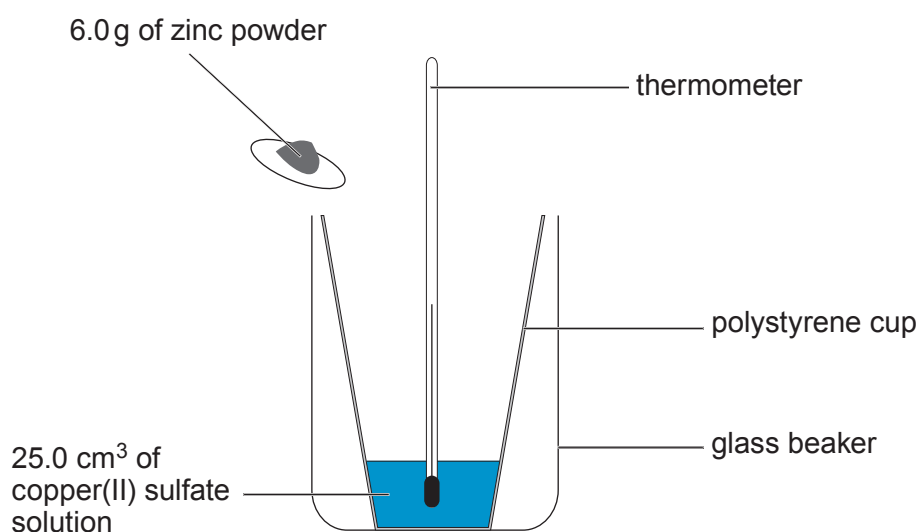
4. (a) The list shows part of the reactivity series.

magnesium
zinc
iron
copper

Underline the correct word(s) in the brackets to complete the following sentence. [1]

The reactivity of metals (**increases** / **stays the same** / **decreases**) down the list.

- (b) Aled investigated the reaction between zinc and excess copper(II) sulfate solution.



Aled put 25.0 cm³ of copper(II) sulfate solution into a polystyrene cup and recorded its temperature. He added 6.0 g of zinc powder to the solution and recorded the temperature every 10 s for 70 s. His results are shown in the table.

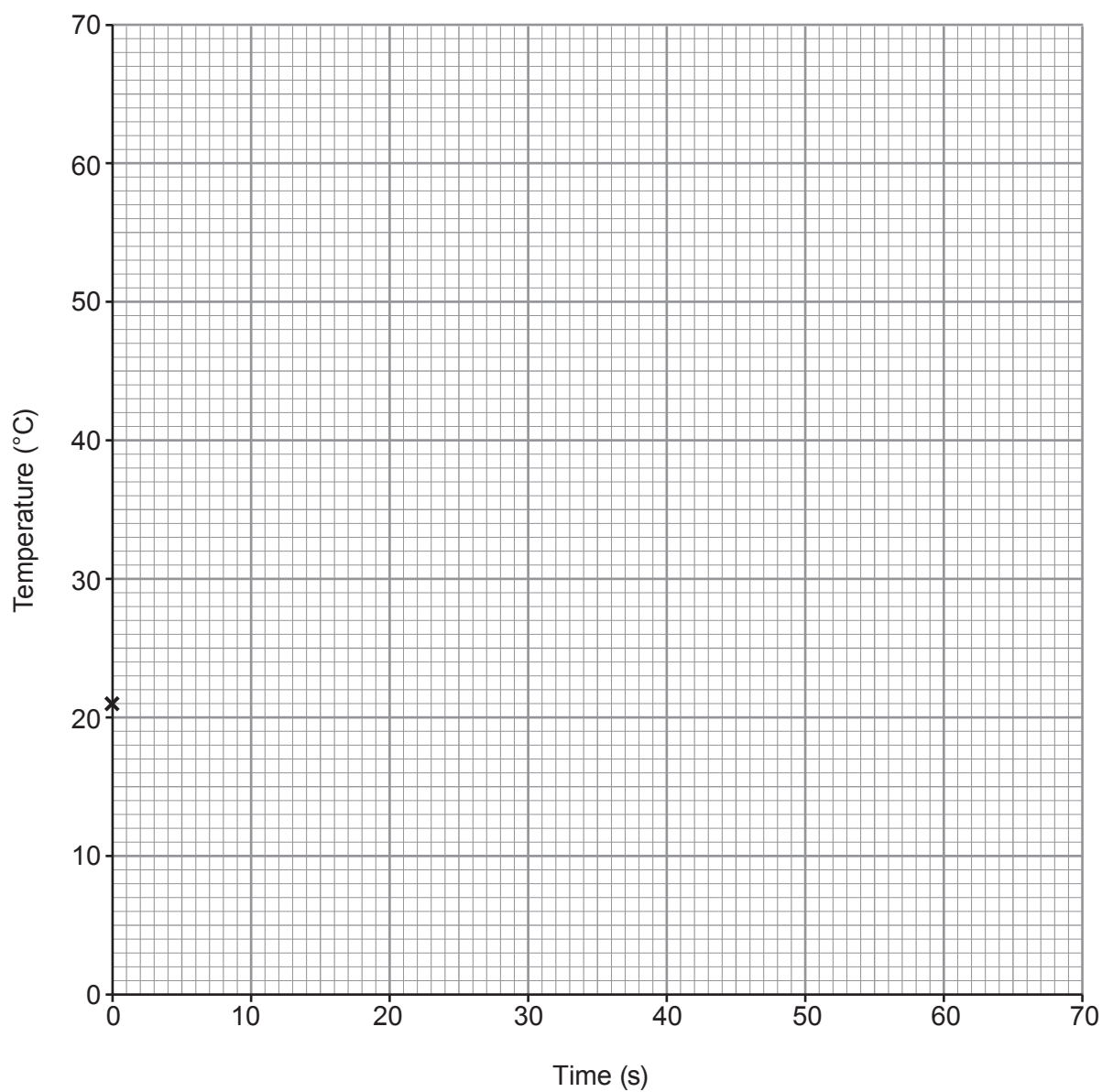
Time (s)	0	10	20	30	40	50	60	70
Temperature (°C)	21	38	50	58	60	59	56	52

- (i) Plot these results on the grid opposite. Draw a suitable line.

The first point has been plotted for you.

[3]





(ii) **Circle** the time it took for **all** the zinc powder to be used up.

[1]

30 s

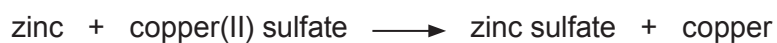
40 s

50 s

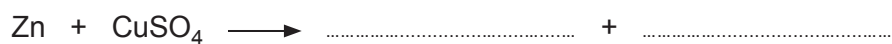
70 s



- (iii) The word equation for the reaction between zinc and copper(II) sulfate solution is as follows.



- I. Complete the symbol equation for the reaction. [2]



- II. Underline the type of reaction taking place. [1]

combustion

displacement

neutralisation

addition



5. (a) The table shows some properties of three plastics.

Plastic	Properties		
polycarbonate	high impact strength	good heat resistance	transparent
bakelite	rigid	good chemical resistance	good thermal insulator
coated nylon	good electrical insulator	flexible	waterproof

Use only the information in the table to answer this part-question.

State which plastic is used for making each of the following items.

Choose the **two** main properties that make the plastic suitable.

Give the reasons why these properties are important. [6 QER]





Examiner
only

Area with horizontal dotted lines for writing.

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(b)

Recycling Plastics

Of all the materials we recycle, plastic is the most complex. The problem is that some plastic packaging is made from several different types of plastic. Different types of plastic are identified by different symbols.

Type of plastic	Symbol	Ease of recycling	Uses
PET		easy	soft drink bottles, rigid food packaging
HDPE		easy	milk cartons, cleaning product bottles, yoghurt pots
PVC		difficult	window frames, guttering
LDPE		difficult	food bags, shopping bags
PP		easy	margarine tubs, microwave meal trays
PS		difficult	some yoghurt pots, takeaway meal trays
other		cannot be recycled	mixed plastic products e.g. salad bags, crisp bags

Waste plastic is collected in different ways.

- Some local authorities collect all their recycling in one bin
- Others ask households to separate their plastics from the rest of their recycling

Sometimes plastic waste cannot be recycled and has to be sent to landfill or incinerated instead. This happens for several reasons.

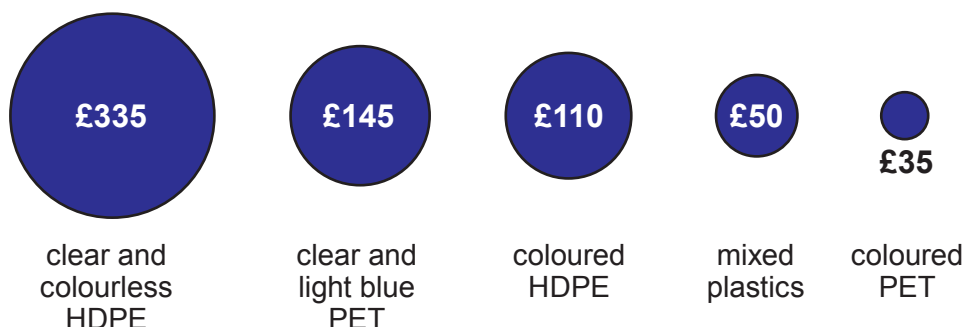
- People are confused about which plastics can and cannot be recycled
- The plastic packaging is contaminated with food waste
- In areas where recycling materials are collected in one bin, one type of waste can contaminate another

Most plastic can be recycled – but it is not always economical to do so.

- Bottles attract the best prices, especially clear ones
- Coloured plastic is less desirable; the colour cannot be removed so its use is limited
- Polystyrene is almost never recycled because there is no market for it



Maximum price per tonne of used bottles (2018)



- (i) Put a **tick (✓)** next to the correct statement.

[1]

all plastics are easy to recycle

☐

PET, HDPE and PP are easy to recycle

☐

PET, HDPE and PVC are difficult to recycle

☐

PVC, LDPE and PS cannot be recycled

☐

- (ii) Put a **tick (✓)** in **two** boxes which explain why some plastic items have to go to landfill.

[2]

the plastic items still contain food waste

☐

not enough plastic items have been collected to make recycling worthwhile

☐

the plastic item is made up of more than one type of plastic

☐

too many plastic items have been collected for recycling

☐

the plastic items are too large to fit in the incinerator

☐

- (iii) **Circle** the type of plastic for which there is the greatest demand for recycling.

[1]

coloured PET

clear and light blue PET

clear and colourless HDPE

coloured HDPE



6. (a) The table shows the names, molecular formulae and structural formulae of some alkanes.

(i) Complete the table.

[3]

Name	Molecular formula	Structural formula
.....	CH ₄	<pre> H H — C — H H </pre>
ethane	C ₂ H ₆	
propane		<pre> H H H H — C — C — C — H H H H </pre>
butane	C ₄ H ₁₀	<pre> H H H H H — C — C — C — C — H H H H H </pre>

- (ii) Give the **names** of the **two** elements present in all alkanes.

[1]

..... and

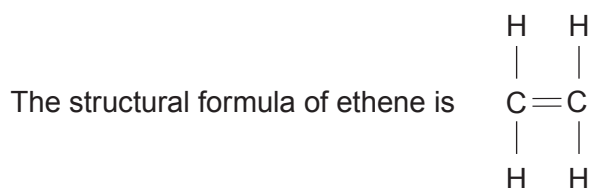
- (iii) The general formula for the alkanes is C_nH_{2n+2}.

Circle the molecular formula for the alkane containing 6 carbon atoms.

[1]

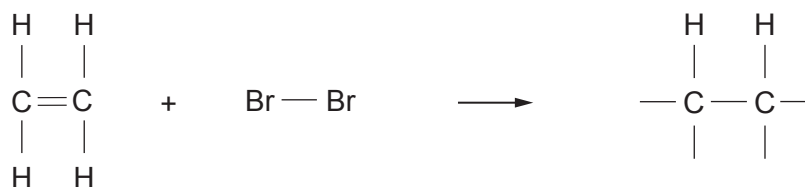


- (b) Ethene belongs to a different family of hydrocarbons called the alkenes.



The presence of the double bond in alkenes can be identified using bromine water.

- (i) Complete the structural formula for the product of the reaction between ethene and bromine. [1]



- (ii) Complete the sentence below by underlining the correct words in the brackets. [2]

Ethene turns (**purple / orange / green / colourless**) bromine water
(**green / milky / colourless / blue**).

- (c) Underline the name given to the type of reaction taking place when ethene reacts with bromine. [1]

addition

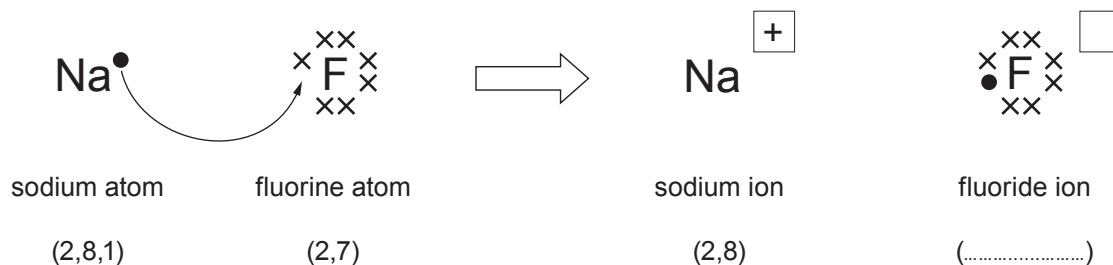
polymerisation

oxidation

neutralisation



7. (a) The diagram shows the electronic changes that occur when sodium reacts with fluorine to form sodium fluoride. The ● and × symbols represent outer shell electrons.

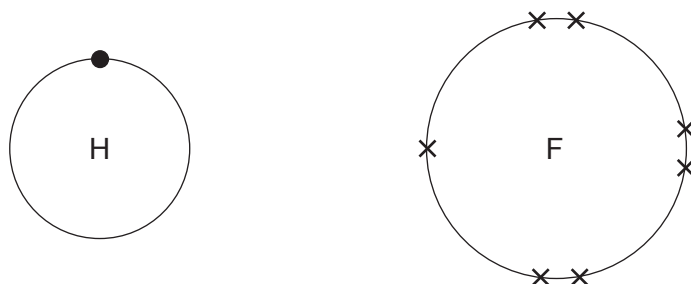


- (i) **Complete the diagram** by writing the electronic structure of the fluoride ion in the brackets and its charge in the box. [2]
- (ii) Complete and balance the equation for the reaction between sodium and fluorine to form sodium fluoride. [2]



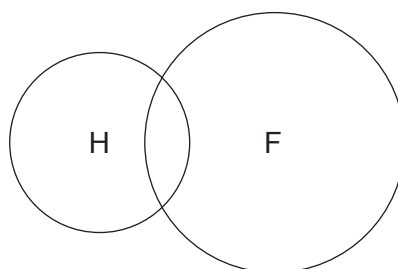
- (b) Hydrogen reacts with fluorine to form hydrogen fluoride, HF.

The diagrams show the outer shell electrons in an atom of hydrogen and an atom of fluorine.

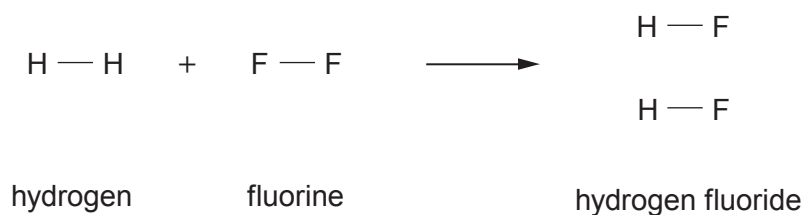


Complete the diagram below to show the outer shell electrons in a molecule of hydrogen fluoride.

[2]



- (c) The equation shows the bonds which are broken and the bonds which are formed when hydrogen reacts with fluorine to form hydrogen fluoride.



The relevant bond energies are shown in the table.

Bond	Bond energy (kJ)
H — H	436
F — F	154
H — F	565

- (i) The total energy needed to break the bonds in the H—H and F—F molecules is 590 kJ. Use the information in the table to show how this value is obtained. [1]
- (ii) Use the information in the table to calculate the total energy released when the bonds in the **two** H—F molecules are formed. [1]

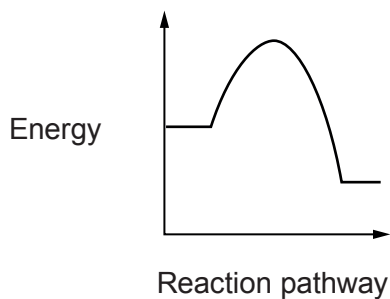
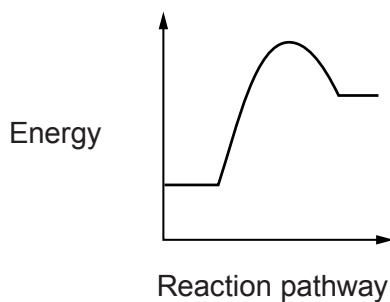
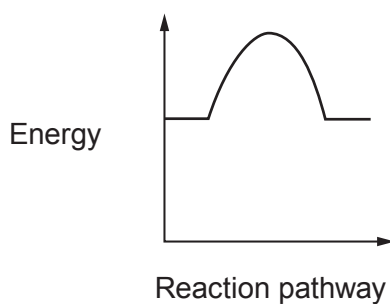
Energy = kJ



- (iii) The formation of hydrogen fluoride is an exothermic reaction.

Put a **tick (✓)** in the box next to the energy profile diagram that shows an exothermic reaction.

[1]

☐☐☐

8. (a) Crude oil can be separated into simpler mixtures called fractions. The table shows information about some of the main fractions obtained from crude oil by fractional distillation.

Fraction	Boiling point range (°C)	Number of carbon atoms present in the alkanes
petroleum gases	< 20	C ₁ –C ₄
petrol	30–75	C ₅ –C ₈
naphtha	70–180	C ₉ –C ₁₂
kerosene	180–250	C ₁₃ –C ₁₆
diesel oil	250–340	C ₁₇ –C ₂₀
lubricating oil	340–500	C ₂₀ –C ₂₄
fuel oil	490–580	C ₂₅ –C ₂₈

- (i) Decane has a boiling point of 174 °C. Give the name of the fraction which contains decane. [1]

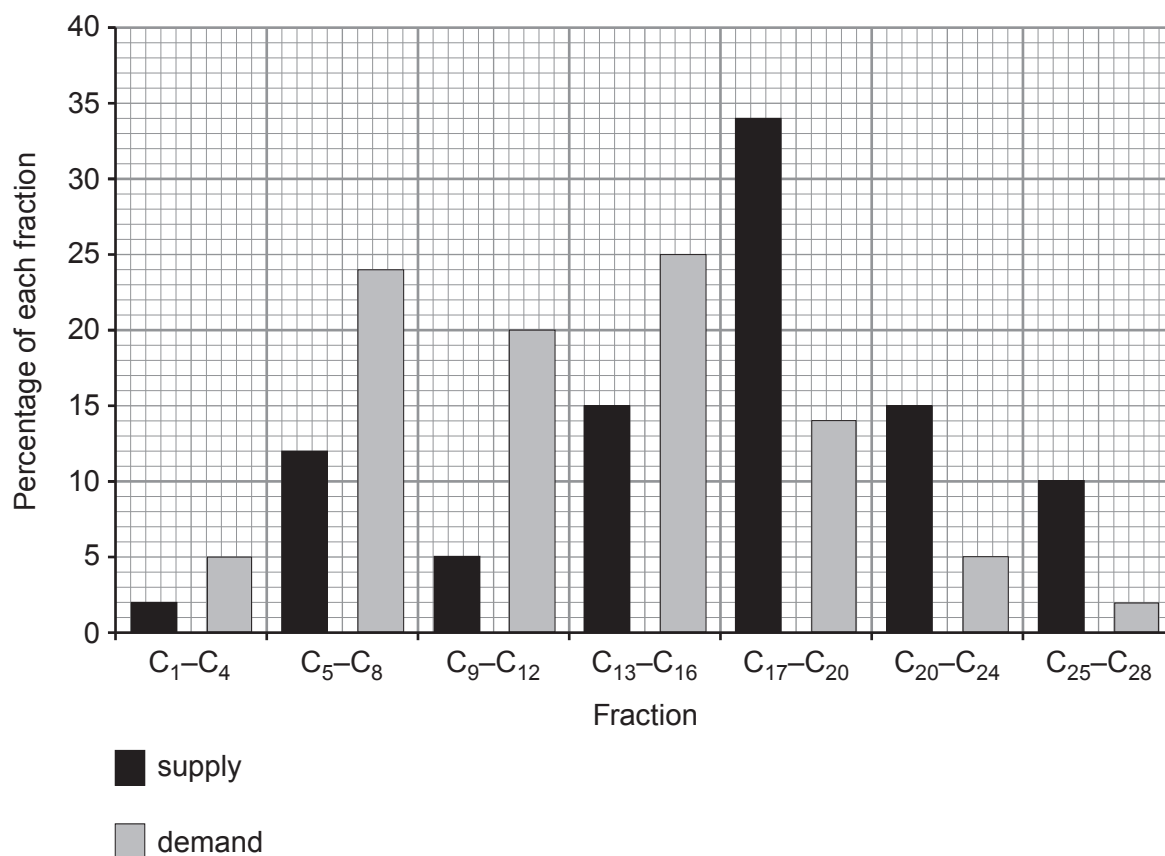
.....

- (ii) One alkane is found in both diesel oil and lubricating oil. Give the number of carbon atoms in this alkane. [1]

.....



- (b) The bar chart shows the supply and demand for some of the fractions obtained from crude oil.



- (i) Give the fraction where the demand is 100 % greater than the supply. [1]

.....

- (ii) Put a **tick** (✓) in the box next to the statement that best describes how the supply and demand of the fractions change as the chain length increases. [1]

supply is greater than demand for all fractions

☐

supply is greater than demand up to C₁₆ after which demand is greater than supply

☐

demand is greater than supply up to C₁₆ after which supply is greater than demand

☐

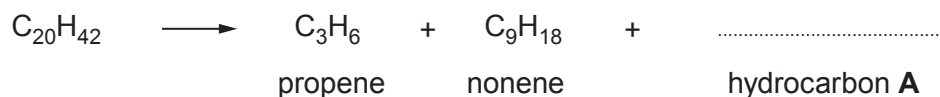
the difference between supply and demand increases up to C₁₆ after which it decreases

☐


- (iii) Oil companies have solved the problem of over-supply of some fractions using a process called cracking.

The alkane $C_{20}H_{42}$ can be cracked forming propene, nonene and **one molecule** of hydrocarbon **A**.

- I. Complete the equation for the cracking of $C_{20}H_{42}$. [1]



- II. Propene is an important raw material in the production of polypropene.

Give the name of the process used to make polypropene from propene. [1]

.....



- (c) Every year thousands of acres of forests are destroyed by wildfires. Fire-fighters use several different methods to put out this type of fire.



State **three** methods that are used to put out forest fires.

Give the part of the fire triangle being removed in each method.

Each method should refer to a different part of the fire triangle.

[3]

Method 1

Part of the fire triangle being removed

Method 2

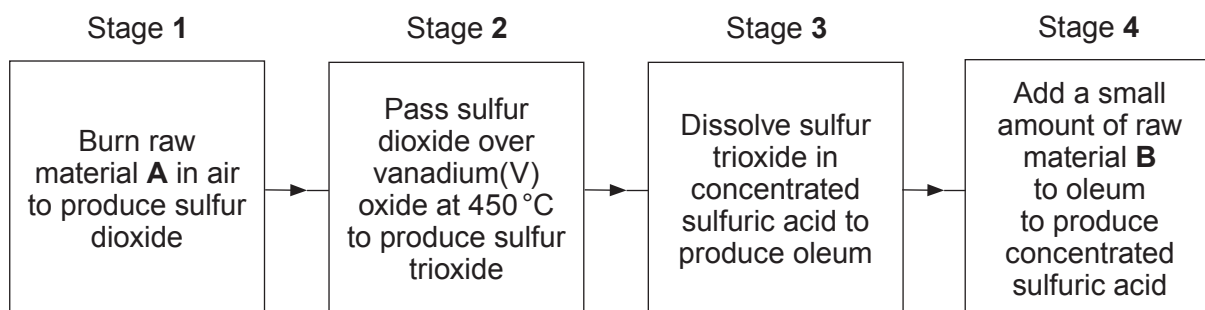
Part of the fire triangle being removed

Method 3

Part of the fire triangle being removed



9. (a) Sulfuric acid is produced by the contact process. The flow diagram shows the main stages in the process.



- (i) Give the name of raw materials **A** and **B**. [2]

A

B

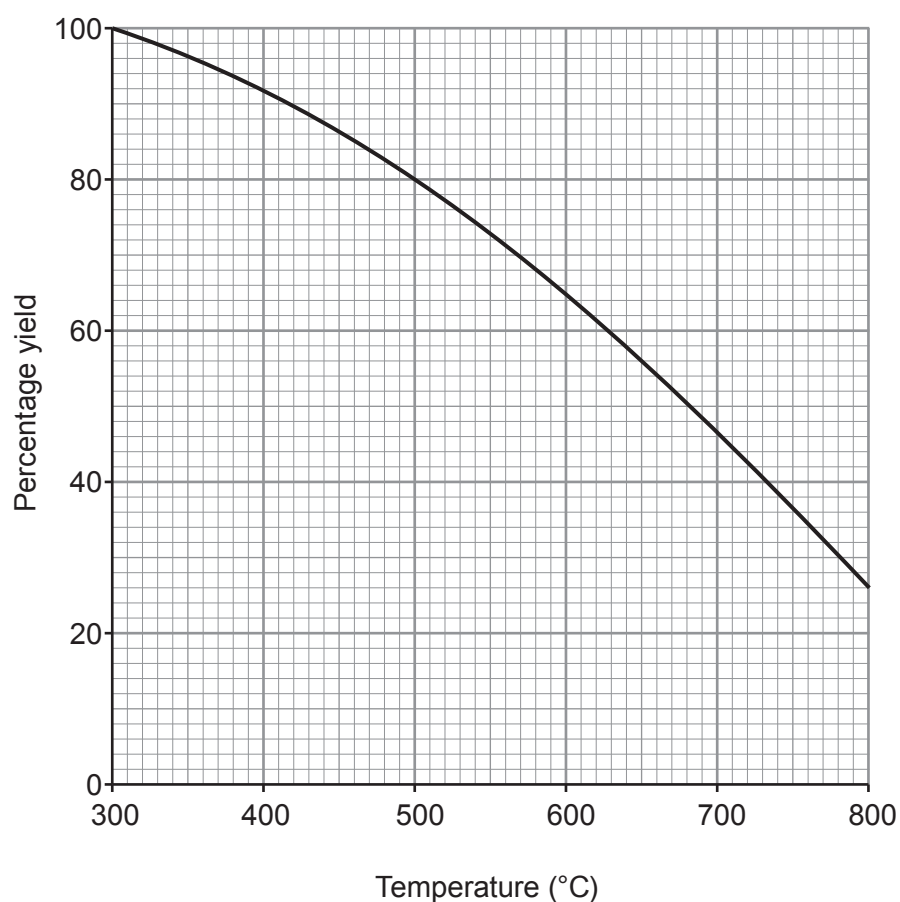
- (ii) State the purpose of vanadium(V) oxide in stage 2. [1]

.....

- (iii) Complete and balance the equation for the reaction in stage 2. [2]



- (iv) The graph shows how the percentage yield of sulfur trioxide changes with temperature between 300 °C and 800 °C.



Use the graph to answer parts I and II.

- I. State the trend in the percentage yield of sulfur trioxide as the temperature increases. [1]

- II. Give the temperature **range** to be used to obtain a yield of sulfur trioxide greater than 80%. [1]

..... to °C

- (v) One molecule of sulfur trioxide reacts with one molecule of sulfuric acid to form one molecule of oleum as the **only product**.

Complete the equation for this reaction by giving the formula of oleum. [1]



- (b) The photograph shows the exothermic reaction between concentrated sulfuric acid and sugar, $C_{12}H_{22}O_{11}$.



- (i) Name the **two** products formed. [2]
..... and
- (ii) State the type of reaction taking place. [1]
.....

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



32

2

Ω

10

0

1 H Hydrogen 1																													
7	Li	9								11	B	12	C	14	N	16	O	19	F	20	Ne								
Lithium	3	Be								Boron	5	Carbon	6	Nitrogen	7	Oxygen	8	Fluorine	9	Neon									
23	Na	24								27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar								
Sodium	11	Mg								Aluminium	13	Silicon	14	Phosphorus	15	Sulfur	16	Chlorine	17	Argon									
39	K	40								70	Ga	73	Ge	75	As	79	Se	80	Br	84	Kr								
Potassium	19	Ca								Gallium	31	Germanium	32	Arsenic	33	Selenium	34	Bromine	35	Krypton									
86	Rb	88								115	In	119	Sn	122	Sb	128	Te	127	I	131	Xe								
Rubidium	37	Sr								Indium	49	Tin	50	Antimony	51	Tellurium	52	Iodine	53	Xenon									
133	Cs	137								204	Tl	207	Pb	209	Bi	210	Po	210	At	222	Rn								
Caesium	55	Ba								Thallium	81	Lead	82	Bismuth	83	Polonium	84	Astatine	85	Radon									
223	Fr	226								201	Hg	207	Pb	209	Bi	210	Po	210	At	222	Rn								
Francium	87	Ra								Mercury	80	Lead	82	Bismuth	83	Polonium	84	Astatine	85	Radon									
Key																													
227																													
Actinium																													
89																													
AC																													
Radium																													
88																													
Ra																													
226																													
Fr																													
223																													
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Rubidium																													
37																													
Y																													
89																													
Yttrium																													
39																													
Zr																													
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Zirconium																													
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Niobium																													
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Molybdenum																													
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Tc																													
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Ru																													
101																													
Ruthenium																													
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Ag																													
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Silver																													
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Cd																													
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Sb																													
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Caesium																													
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Ba																													
137																													
Barium																													
57																													
La																													
139																													
Lanthanum																													
72																													
Hf																													
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Pt																													
195																													
Platinum																													
78																													
Au																													
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Gold																													
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Hg																													
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Thallium																													
81																													
Pb																													
207																													
Lead																													
82																													
Bi																													
209																													
Bismuth																													
83																													
Po																													
210																													
Polonium																													
84																													
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Radon																													
86																													
Fr																													
223																													
Francium																													
87																													
Ra																													
226																													
Radium																													
88																													
Ac																													
227																													
Actinium																													
89																													

Key

relative atomic mass

A_r	Symbol	Name	Z
-------	--------	------	-----

Name	atomic number
7	

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE**

3410U20-1



Z22-3410U20-1

FRIDAY, 27 MAY 2022 – MORNING

CHEMISTRY – Unit 2:
Chemical Bonding, Application of Chemical Reactions
and Organic Chemistry

FOUNDATION TIER

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	7	
3.	7	
4.	13	
5.	10	
6.	6	
7.	8	
8.	6	
9.	7	
10.	7	
Total	80	

3410U201
01

JUN223410U20101

Answer **all** questions.

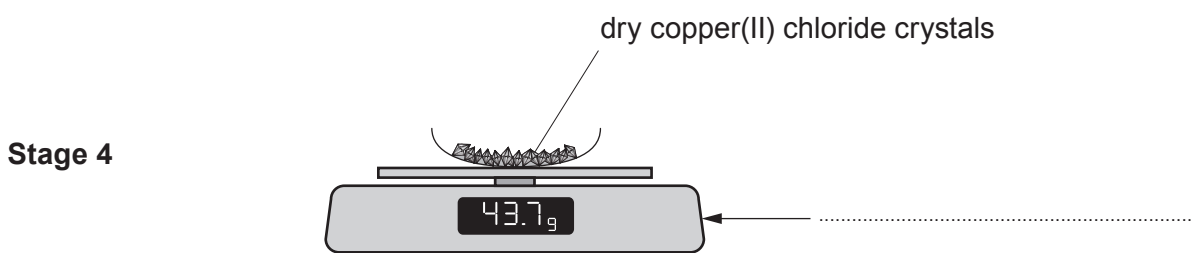
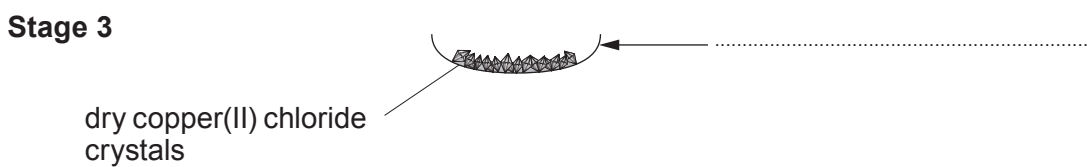
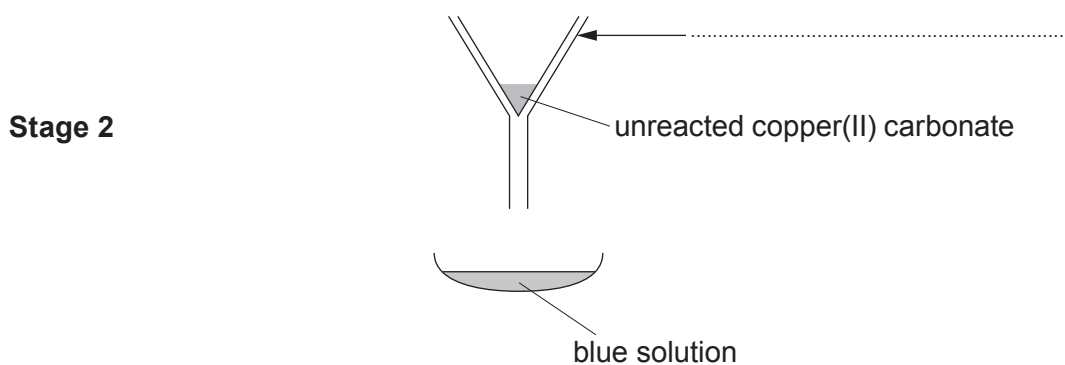
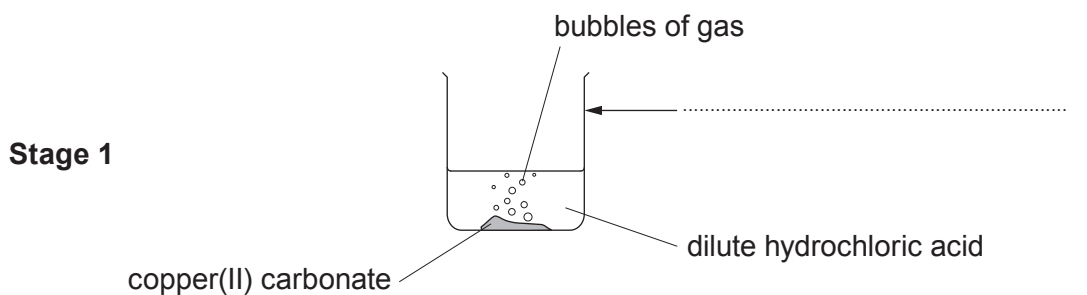
1. Selwyn carried out an experiment to prepare copper(II) chloride crystals.

The diagrams show the stages of the experiment he carried out.

- (a) Choose apparatus from the box to label the diagrams.

[4]

evaporating basin	conical flask	filter funnel	electronic balance
filter paper	beaker	test tube	



- (b) In **Stage 1** Selwyn added copper(II) carbonate until **all** the dilute hydrochloric acid was used up.

Tick (✓) the box next to the statement which best describes what Selwyn would see when all the acid had been used up. [1]

bubbling increases

☐

bubbling stops

☐

bubbling decreases

☐

- (c) The gas formed in **Stage 1** turns limewater milky. Underline the name of this gas. [1]

oxygen

hydrogen

carbon dioxide

nitrogen

- (d) Choose words from the box to complete the following sentences. [2]

evaporation

filtration

distillation

neutralisation

The process used to remove the unreacted copper(II) carbonate in **Stage 2** is called

.....

The process used to remove water in **Stage 3** is called

- (e) The container holding the crystals has a mass of 29.8 g. Using the information given in **Stage 4**, calculate the mass of the crystals formed. [1]

Mass of crystals formed = g



2. (a) Crude oil is formed from simple marine organisms.

Tick (✓) the box next to the length of time it takes for the organisms to turn into crude oil.

[1]

hundreds of years

☐

thousands of years

☐

millions of years

☐

- (b) Crude oil can be separated into simpler mixtures called fractions.

Tick (✓) the box next to the method used to separate crude oil into fractions.

[1]

fractional distillation

☐

filtration

☐

cracking

☐

polymerisation

☐


(c) The table shows properties of some fractions that are obtained from crude oil.

Fraction	Size of molecules (chain length)	Viscosity	Ease of ignition	Amount of smoke formed
petrol	$C_5 - C_{10}$	very runny	very easy	no smoke
naphtha	$C_8 - C_{12}$	fairly runny	quite easy	little smoke
kerosene	$C_{10} - C_{16}$	thick	quite hard	quite a lot of smoke
diesel oil	$C_{14} - C_{20}$	very thick	very hard	very smoky

Use only the information from the table to answer the following questions.

(i) Name the fraction which is the easiest to pour. [1]

(ii) Name the fraction which is the hardest to burn. [1]

(iii) Name the fraction with the smallest **range** of chain lengths. [1]

.....

(iv) Name the fraction which burns with the cleanest flame. Give the reason for your choice. [2]

Fraction

Reason

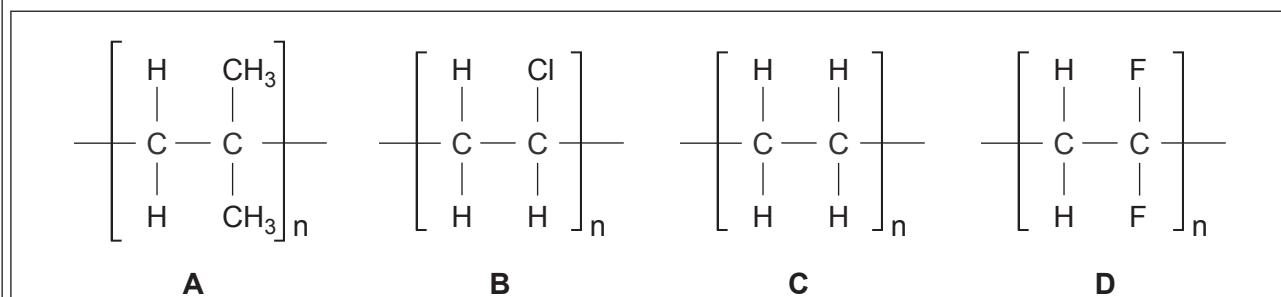


3. (a) Complete the table.

[2]

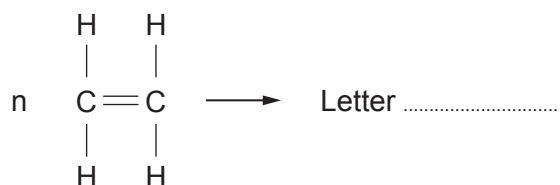
Name	Molecular formula	Structural formula
ethane	C_2H_6	<pre> H H H — C — C — H H H </pre>
propane	C_3H_8	
butane		<pre> H H H H H — C — C — C — C — H H H H H </pre>

(b) The box contains four polymers **A**, **B**, **C** and **D**.

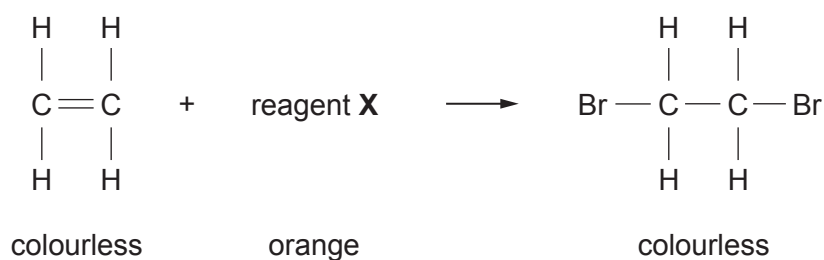


Give the **letter** of the polymer formed during the polymerisation of ethene.

[1]



(c) The equation shows the reaction taking place when testing for a $C=C$ bond.



bromine water

potassium dichromate

universal indicator

barium chloride

Choose the name of reagent **X** from the box above.

.....

[1]

(d) The equation shows the fermentation of glucose.



(i) Underline the chemical name for $\text{C}_2\text{H}_5\text{OH}$.

[1]

sugar

biofuel

ethanol

alcohol

(ii) Calculate the relative molecular mass, M_r , of $\text{C}_2\text{H}_5\text{OH}$.

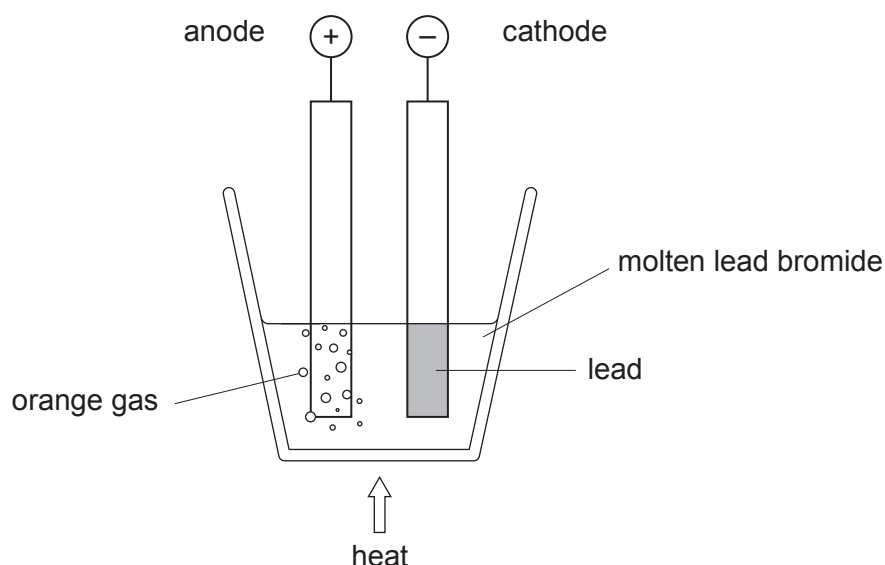
[2]

$$A_r(\text{H}) = 1 \quad A_r(\text{O}) = 16 \quad A_r(\text{C}) = 12$$

$$M_r = \dots\dots\dots$$



4. (a) The diagram shows the apparatus a teacher used to demonstrate what happens when an electric current is passed through molten lead bromide.



- (i) Lead bromide contains the ions Pb^{2+} and Br^- .

Underline the correct formula of lead bromide.

[1]

Pb_2Br PbBr PbBr^2 PbBr_2

- (ii) Give the state (solid, liquid or gas) of the lead bromide during the process.

[1]

.....

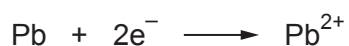
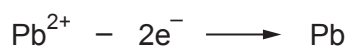
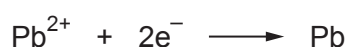
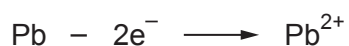
- (iii) Name the orange gas formed at the anode.

[1]

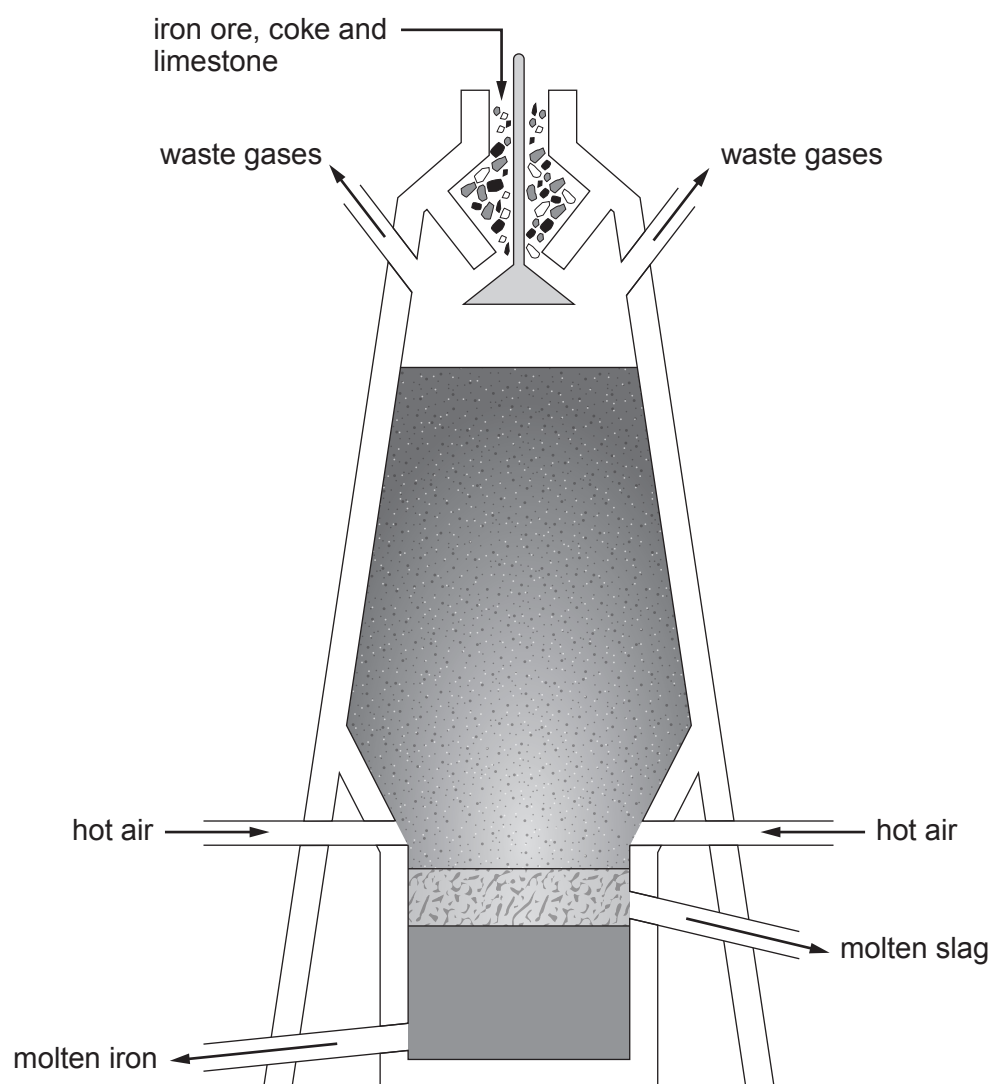
- (iv) During the process each lead ion, Pb^{2+} , gains two electrons forming a lead atom at the cathode.

[1]

Tick (✓) the box next to the equation for the reaction at the cathode.


☐

☐

☐

☐


- (b) Iron is extracted from its ore in the blast furnace. The diagram shows the materials which enter and leave the furnace.



- (i) Underline the correct word(s) in the brackets to complete each sentence. [3]

The furnace is heated by burning (**coke**/**iron**/**iron ore**).

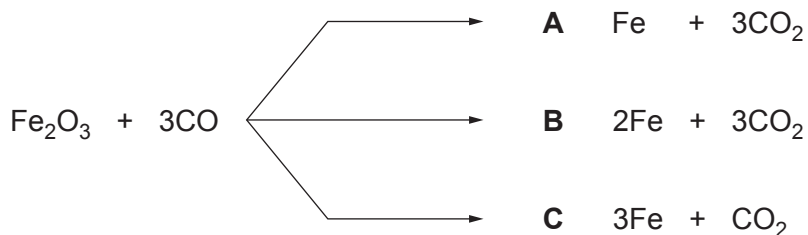
Hot air provides (**waste gases**/**oxygen**/**slag**) for burning to take place.

Impurities are removed by adding (**steel**/**hot air**/**limestone**) to the furnace.

- (ii) The word equation shows the main reaction taking place in the blast furnace.



Give the **letter** next to the products which correctly balance the symbol equation for this reaction. [1]



Letter



(c) Chemists have designed a wide variety of alloys for different uses.

Some alloys contain iron and carbon only, whereas others contain additional metals.

The table shows the composition and properties of some alloys.

Alloy	Composition	Properties
mild steel	iron plus 0.15–0.30 % carbon	malleable (easy to bend), ductile (easy to pull into wire) and soft (easy to scratch)
high carbon steel	iron plus 0.70–1.50 % carbon	strong, brittle and hard
cast iron	iron plus 2.00–5.00 % carbon	very strong, very brittle and very hard

Use only the information in the table above to answer parts (i)–(iii).

(i) Name the alloy that contains the **least** amount of carbon. [1]

.....

(ii) Underline the effect of increasing the percentage (%) of carbon in these alloys. [1]

strength decreases

hardness decreases

softness increases

brittleness increases

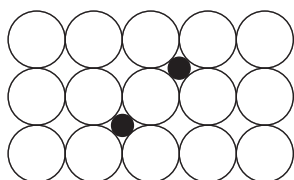
(iii) Give the property of mild steel which makes it useful for making car bodies. [1]

.....

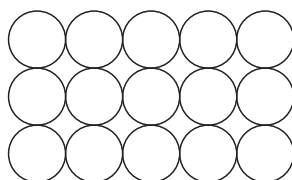


- (iv) Diagrams **A**, **B** and **C** are models showing the arrangement of atoms in pure iron and in two alloys, **but not necessarily in that order**.

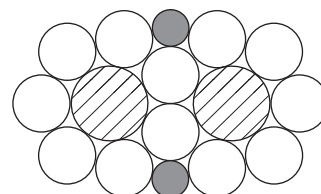
Choose the **letter** of the model which best represents cast iron. Give the reason for your choice. [2]



A



B



C

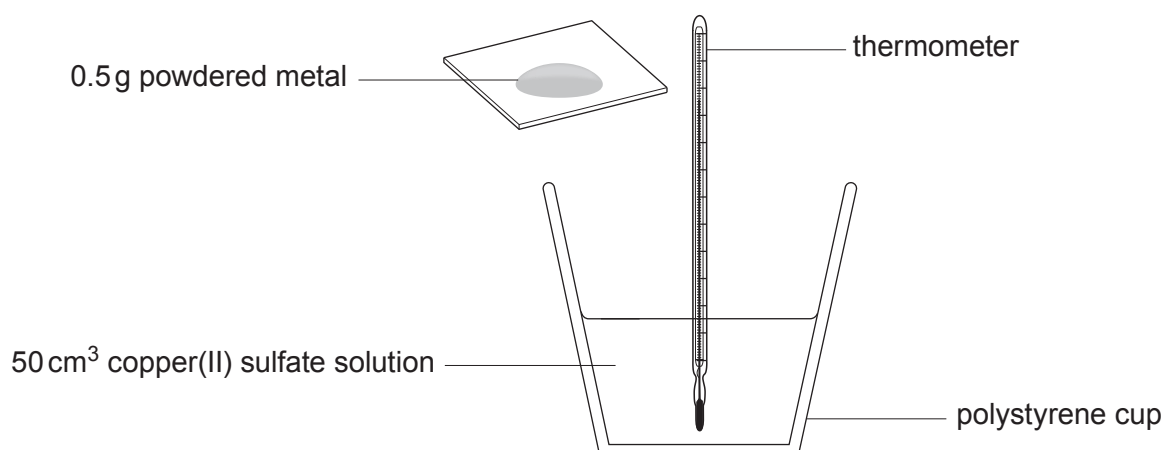
Letter

Reason



5. Joe, Alex and Megan were asked to investigate the temperature rise when four metals were added to excess copper(II) sulfate solution. 0.5 g of each metal was added to separate 50 cm³ samples of copper(II) sulfate solution.

The temperature rise for each reaction was measured.

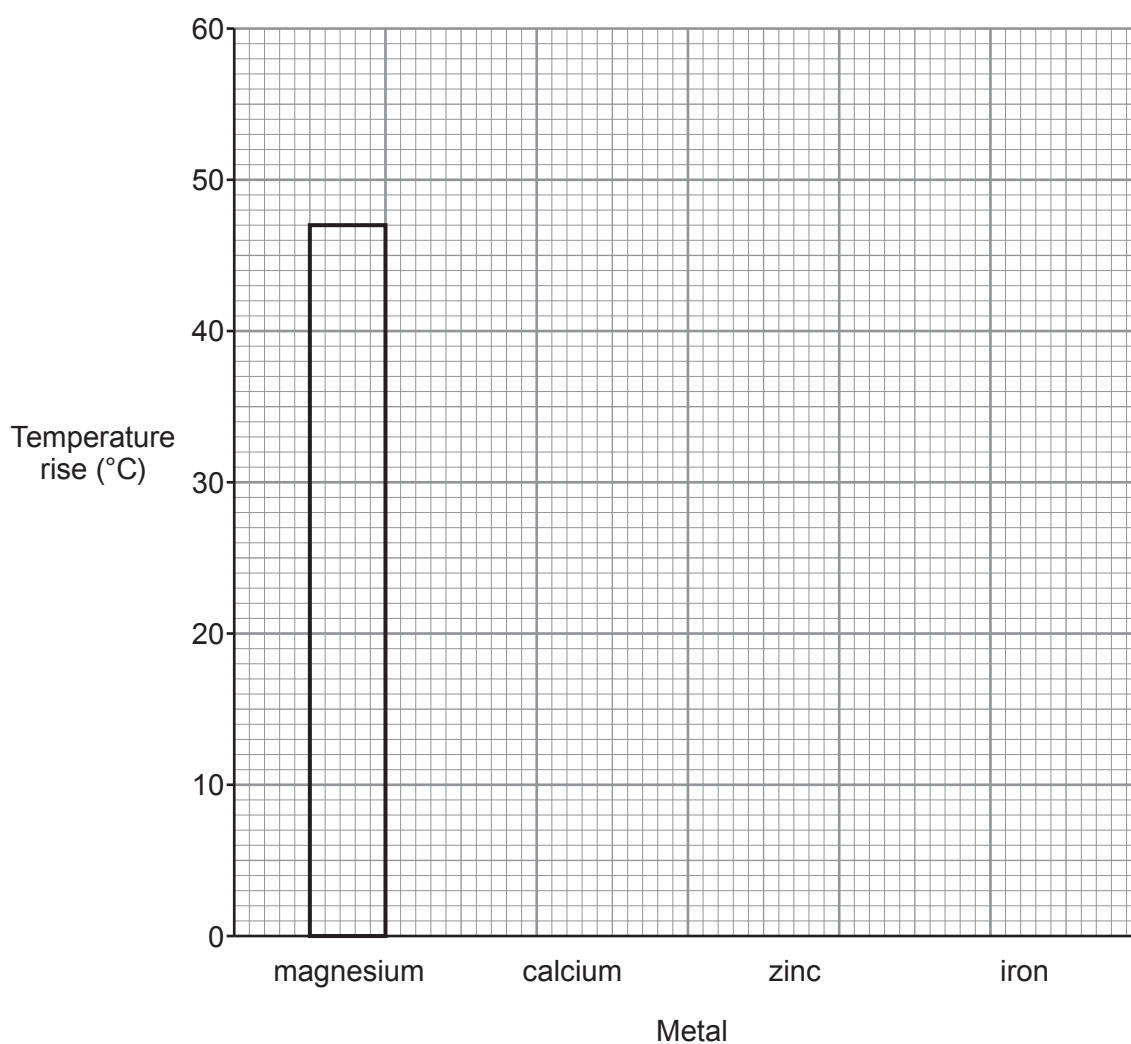


Their results are shown in the table below.

Metal	Temperature rise (°C)
magnesium	47
calcium	54
zinc	38
iron	25



- (a) (i) Draw a bar chart on the grid to show the results. One bar has been drawn for you. [2]



- (ii) Put the **four** metals in order of reactivity. [1]

Most reactive

.....

.....

Least reactive



- (iii) The temperature rise for each metal was lower than expected. The students were asked to suggest how the apparatus could be improved so that the temperature rise recorded for each metal was closer to the expected value.

Joe suggested wrapping the polystyrene cup in cotton wool.

Megan suggested putting a lid on the polystyrene cup.

Alex suggested using a cup made from copper.

Choose which student's suggestion would **not** result in the temperature rises being closer to the expected value. Explain your choice.

[2]

Student

Reason

.....

.....

- (iv) Complete the equation for the reaction between magnesium and copper(II) sulfate solution.

[2]



- (b) The students repeated the experiment using chromium and recorded a temperature rise of 30 °C.

What does the temperature rise of 30 °C tell you about the reactivity of chromium compared to the four metals in part (a)?

[1]

.....

.....

- (c) The energy given out by the reaction can be calculated using the formula below.

$$\text{energy given out (J)} = \text{volume of solution (cm}^3\text{)} \times 4.2 \times \text{temperature rise (}^\circ\text{C)}$$

Calculate the energy given out during the displacement reaction between **iron** and copper(II) sulfate solution.

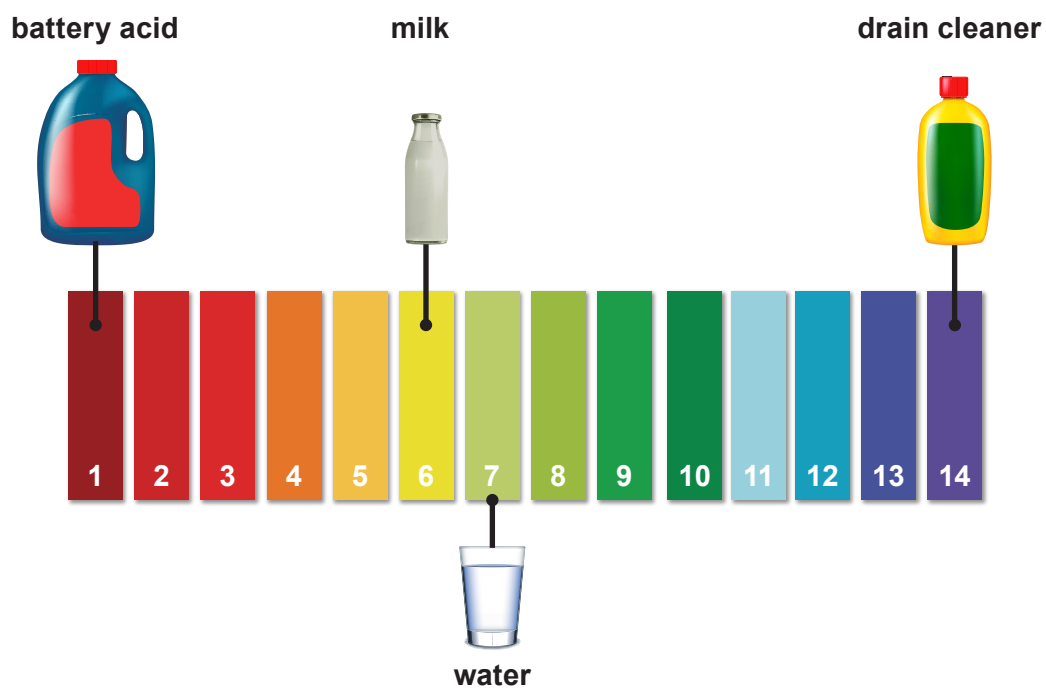
[2]

Energy given out =J

10



6. The picture below shows the universal indicator pH colour chart and the position of some substances on the pH scale.



Examiner
only

Describe what is meant by the pH scale and what it tells us about the substances shown in the picture. [6 QER]

.....

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.....

.....

6



7. (a) Skiers and mountaineers use hand warmers in very cold weather conditions. Hand warmers are small packets that produce heat when needed. Some ski clothing is specially designed to hold a hand warmer.



Table 1 describes how three types of hand warmers work.

A Air-activated (disposable)	B Battery powered (reusable)	C Supersaturated solution (reusable)
The packaging seal is broken allowing air to reach the chemicals causing a chemical reaction to occur. This type of hand warmer can only be used once. 	A metal coil heats up when the device is switched on. The device needs recharging. 	A metal button on the packaging is pressed. This causes crystals to form in the solution. The hand warmer can be reactivated by placing it in boiling water. 

Table 1



Table 2 shows some information about each type of hand warmer.

Type of hand warmer	Cost (£)	Time to warm up after activation	Temperature after being activated and placed in a freezer (°C)				
			after 15 mins	after 30 mins	after 60 mins	after 90 mins	after 120 mins
A	1	less than 1 min	39	39	38	38	37
B	80	less than 1 min	32	29	28	27	26
C	24	less than 1 min	42	34	27	18	8

Table 2

- (i) Tick (✓) the box next to the correct statement. [1]

Gloves need to be worn when using hand warmers

☐

Boiling water is used to recharge battery powered hand warmers

☐

Some chemical reactions give out heat energy

☐

All hand warmers are reusable

☐

- (ii) Give **two** reasons why hand warmer **A** is the most popular choice. [2]

Reason 1

.....

Reason 2

.....



- (b) An air-activated hand warmer contains several chemicals mixed together. One of these chemicals is iron.

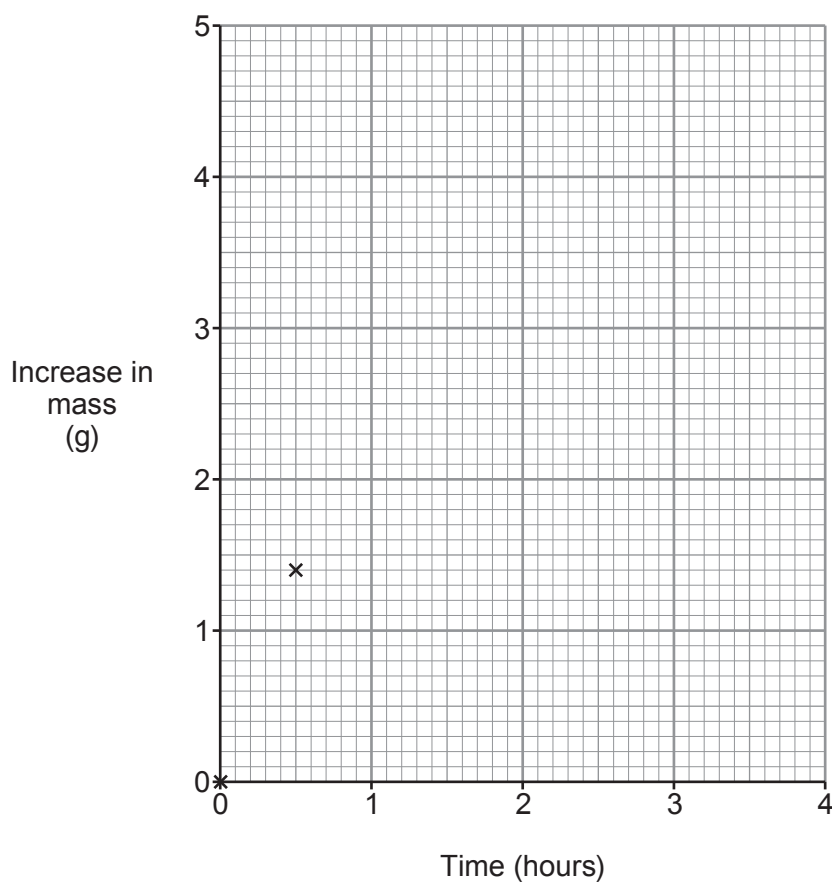
When the pack is opened, a reaction takes place between iron and oxygen in the air, causing an increase in mass.

A student investigated the increase in mass of the opened pack over several hours. The results are shown below.

Time (hours)	Increase in mass (g)
0.0	0.0
0.5	1.4
1.0	2.6
1.5	3.7
2.0	4.4
2.5	4.7
3.0	4.8
3.5	4.8
4.0	4.8



- (i) Plot the mass increase of the pack against time on the grid and draw a suitable line. Two plots have been done for you. [3]



- (ii) Tick (✓) the box next to the time it takes for the reaction to finish. [1]

2 hours ☐ 3 hours ☐ 4 hours ☐ 5 hours ☐

- (iii) Tick (✓) the box next to the statement that best explains the shape of the graph. [1]

Iron reacts with oxygen forming iron oxide until all the oxygen is used up

☐

Heat formed expands the iron

☐

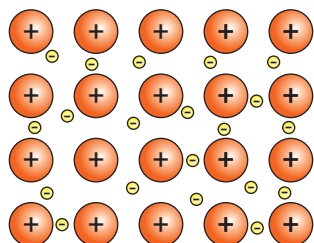
Iron oxide loses oxygen, forming iron

☐

Iron reacts with oxygen forming iron oxide until all the iron is used up

☐

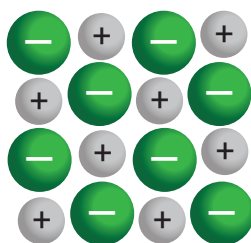

- (iii) The diagrams show four different structures. Give the **letter** of the structure most likely to represent potassium fluoride. [1]



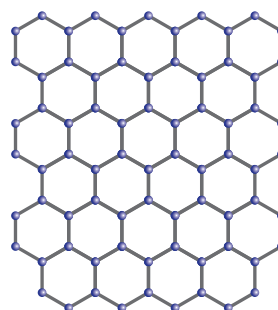
A



B



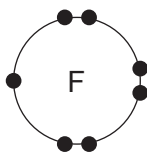
C



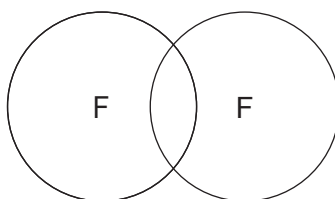
D

Letter

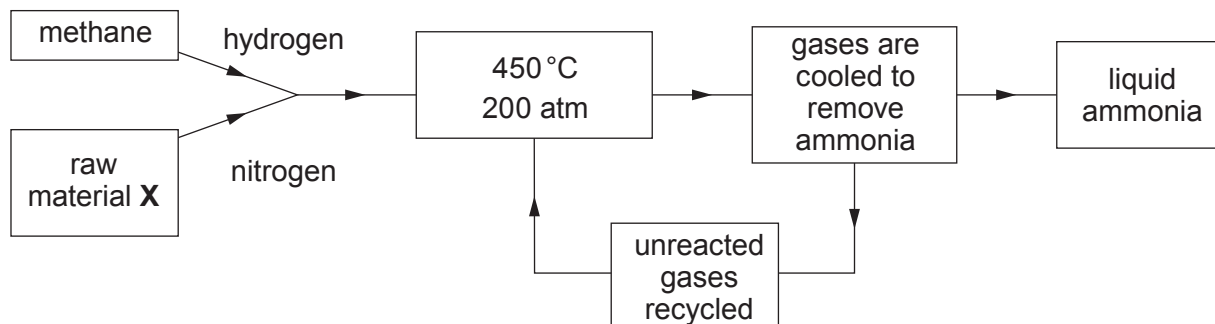
- (b) The diagram shows the electrons in the outer shell of an atom of fluorine.



Complete the diagram to show the outer shell electrons in a molecule of fluorine. [2]



9. (a) The diagram outlines the manufacture of ammonia by the Haber process.



- (i) Name the raw material **X**. [1]
- (ii) The pressure used in the Haber process is 200 atm. State why a higher pressure is **not** used. [1]
-
- (iii) At 450 °C, the reaction is very slow. Iron is used in the process to speed up the reaction. Give the name for a substance used to speed up a chemical reaction. [1]
-
- (iv) The reaction between nitrogen and hydrogen is represented by the equation below.



Complete the equation below using the key:

[2]



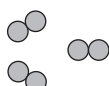
nitrogen gas, N_2



hydrogen gas, H_2



+



- (b) One of the main uses of ammonia is in the manufacture of fertilisers.

The table shows the results obtained when tests were carried out on three different fertilisers **A**, **B** and **C**.

Fertiliser	Test for positive ion	Test for negative ion
A	On adding sodium hydroxide solution and warming, a pungent smelling gas is formed which turns red litmus blue	On adding barium chloride solution a white precipitate forms
B	Lilac flame test	On adding silver nitrate solution a white precipitate forms
C	On adding sodium hydroxide solution and warming, a pungent smelling gas is formed which turns red litmus blue	On adding silver nitrate solution a white precipitate forms

Give the **letter** of the fertiliser which is ammonium sulfate.

[1]

Letter

- (c) Ammonia reacts with chlorine to form nitrogen and hydrogen chloride.

Complete the balancing of the equation for this reaction.

[1]



10. (a) The list below shows part of the reactivity series.

sodium
aluminium
(carbon)
tin
copper
silver

- (i) Tin is extracted from its ore by heating with carbon. Aluminium is extracted from its ore using a different method. Give the name of the method used to extract aluminium. [1]

.....

- (ii) The equation shows the extraction of tin from tin oxide using carbon.



Tick (✓) the box next to the correct statement. [1]

Carbon is reduced

☐

Tin is oxidised

☐

Tin oxide is reduced

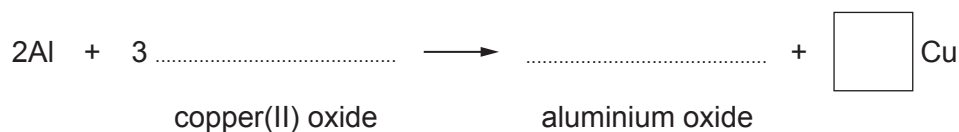
☐

Carbon dioxide is oxidised

☐

- (iii) When aluminium and copper(II) oxide are heated together, aluminium oxide and copper are formed.

Complete and balance the equation for this reaction. [3]



- (b) A teacher wanted to find out the position of four metals **A**, **B**, **C** and **D** in the reactivity series.

She heated each metal in turn with oxides of the other three. The results were as follows.

A reduced the oxide of **C**

B reduced the oxide of **A**

B reduced the oxide of **C**

D reduced the oxide of **B**

Place the metals in order of reactivity.

[2]

Most reactive

.....

.....

Least reactive

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1																				4 He Helium 2															
		9 Be Beryllium 4																		16 O Oxygen 8		19 F Fluorine 9		20 Ne Neon 10											
23 Na Sodium 11		24 Mg Magnesium 12																		32 S Sulfur 16		35.5 Cl Chlorine 17		40 Ar Argon 18											
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21		48 Ti Titanium 22		51 V Vanadium 23		52 Cr Chromium 24		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		59 Ni Nickel 28		63.5 Cu Copper 29		65 Zn Zinc 30		70 Ga Gallium 31		73 Ge Germanium 32		75 As Arsenic 33		79 Se Selenium 34		80 Br Bromine 35		84 Kr Krypton 36	
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39		91 Zr Zirconium 40		93 Nb Niobium 41		96 Mo Molybdenum 42		99 Tc Technetium 43		101 Ru Ruthenium 44		103 Rh Rhodium 45		106 Pd Palladium 46		108 Ag Silver 47		112 Cd Cadmium 48		115 In Indium 49		119 Sn Tin 50		122 Sb Antimony 51		127 I Iodine 53		131 Xe Xenon 54			
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57		179 Hf Hafnium 72		181 Ta Tantalum 73		184 W Tungsten 74		186 Re Rhenium 75		190 Os Osmium 76		192 Ir Iridium 77		195 Pt Platinum 78		197 Au Gold 79		201 Hg Mercury 80		204 Tl Thallium 81		207 Pb Lead 82		209 Bi Bismuth 83		210 Po Polonium 84		210 At Astatine 85		222 Rn Radon 86	
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																															

Key

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE**

3410U20-1



S23-3410U20-1

MONDAY, 22 MAY 2023 – MORNING**CHEMISTRY – Unit 2:****Chemical Bonding, Application of Chemical Reactions
and Organic Chemistry****FOUNDATION TIER**

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 4(b) is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	8	
3.	9	
4.	12	
5.	12	
6.	10	
7.	9	
8.	11	
Total	80	

3410U201
01

JUN233410U20101

Answer **all** questions.

1. (a) A student investigated the pH of several substances using universal indicator solution. The colour produced with each substance is shown.

Substance	Colour
hand wash	blue
battery fluid	red
water	green
lemon juice	orange
drain cleaner	purple

- (i) Name **one** substance in the table which is an alkali. [1]

.....

- (ii) State which substance is the strongest acid. [1]

.....

- (iii) Drain cleaner contains sodium hydroxide. [1]
Circle the correct formula of sodium hydroxide.

NaO

NAOH

NaOH

NaOh



- (b) The student carried out some reactions using hydrochloric acid. She recorded the following observations.

Reaction	Reactant added to acid	Observations
A	magnesium	fizzing temperature increase of 25 °C magnesium disappears
B	sodium hydroxide	no fizzing temperature increase of 8 °C
C	sodium carbonate	fizzing temperature increase of 5 °C
D	copper(II) oxide	no fizzing temperature increase of 11 °C mixture turns blue

- (i) Reaction **A** produced a gas. Give the **letter** of the other reaction that produced a gas. Which observation shows that a gas was produced? [2]

Letter

Observation

- (ii) One of the gases produced is hydrogen.

Describe how you could test for hydrogen gas. Give the observation you would expect for a positive test. [2]

Test

Observation

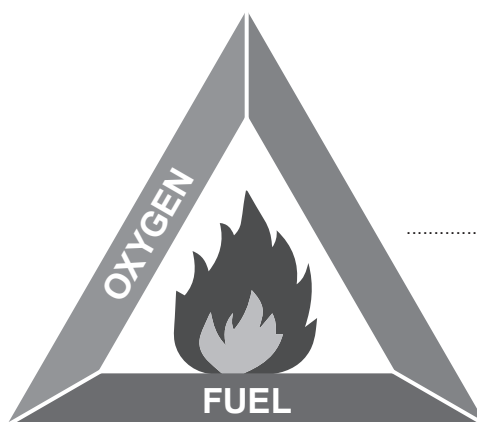
- (iii) Give the **letter** of the reaction which was the **least** exothermic. Give the reason for your answer. [2]

Letter

Reason



2. (a) In the diagram of the fire triangle, two sides have been labelled.



- (i) **Complete the fire triangle** by labelling the third side. [1]

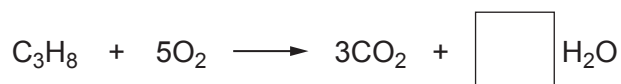
- (ii) Complete the following sentences to explain how a fire is extinguished by the following methods. [2]

Using a fire blanket removes the from the fire triangle.

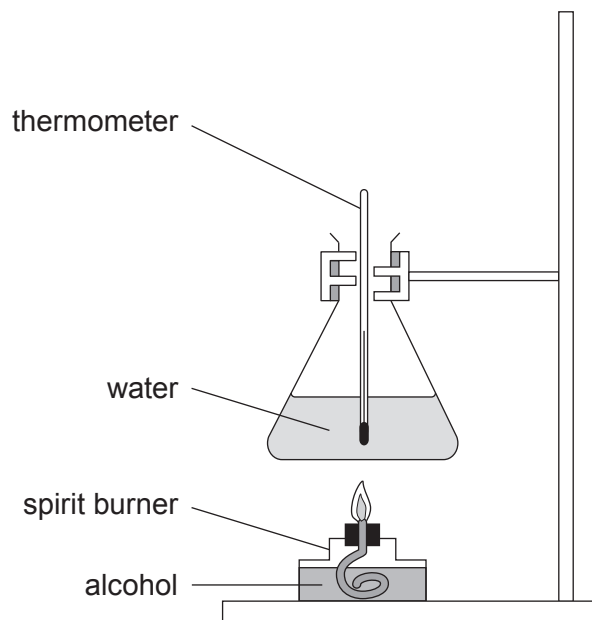
Cutting down trees in a forest fire removes the from the fire triangle.

- (b) Choose a **number** from the box below to balance the equation for the burning of propane gas. [1]

2	3	4	8
---	---	---	---



- (c) A class used the apparatus shown to compare the combustion of different alcohols. They burned each alcohol for 1 minute. They measured the temperature of the water before and after burning each alcohol.



The table shows the increase in temperature of the water for each alcohol.

Alcohol	Temperature increase ($^{\circ}\text{C}$)
methanol	8
ethanol	19
propanol	23
butanol	38

- (i) The starting temperature of the water each time was 18°C . Calculate the final temperature of the water after burning ethanol. [1]

Final temperature = $^{\circ}\text{C}$



(ii) Tick (✓) the question that the class were trying to answer.

[1]

Which alcohol gives out the most heat energy?	
Which gases are produced when alcohols burn?	
Which alcohol has the lowest boiling point?	
Which alcohol burns for the longest?	

(d) Methanol has the chemical formula CH_3OH .

Calculate the relative molecular mass, M_r , of methanol.

[2]

$$A_r(\text{C}) = 12$$

$$A_r(\text{H}) = 1$$

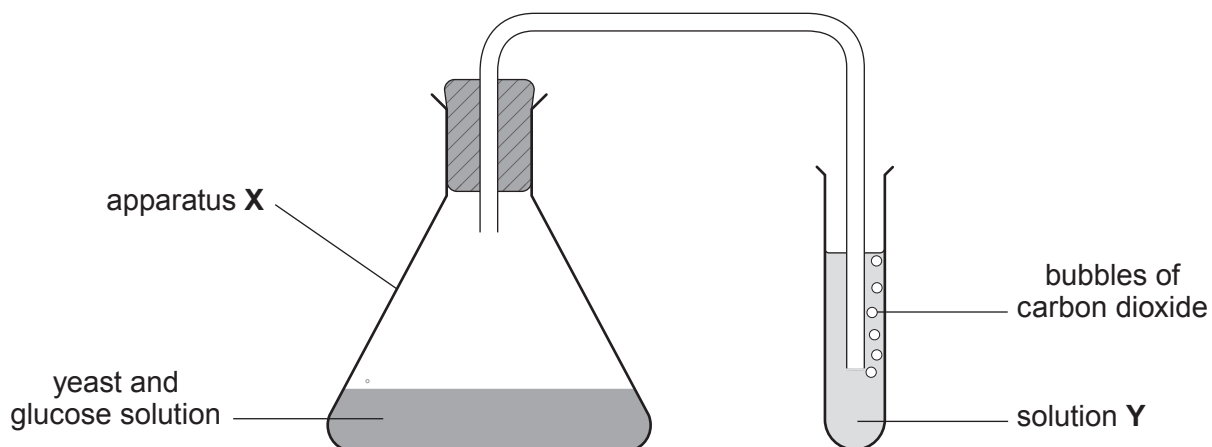
$$A_r(\text{O}) = 16$$

$M_r =$

8



3. (a) Ethanol can be produced by fermentation using the apparatus shown.



- (i) Circle the name of apparatus **X**. [1]

conical flask

measuring cylinder

beaker

- (ii) Solution **Y** can be used to show the presence of carbon dioxide gas.

Give the name of solution **Y** and state what is seen when carbon dioxide is bubbled through it. [2]

Name of solution **Y**

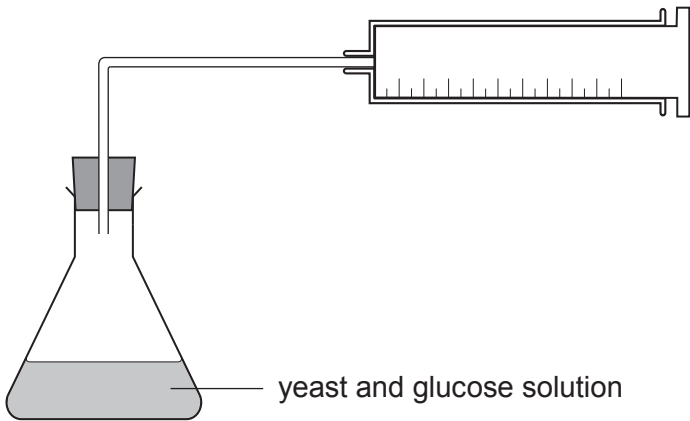
What is seen

- (iii) The chosen temperature for this reaction is 35 °C. Tick (✓) the correct reason why carbon dioxide would **not** be produced at a temperature of 90 °C. [1]

the reaction is finished	
the yeast is used up	
the enzymes in the yeast are denatured	



(b) In a different experiment, using the apparatus shown below, the volume of carbon dioxide produced at 35 °C was measured and recorded every 10 minutes for 60 minutes.



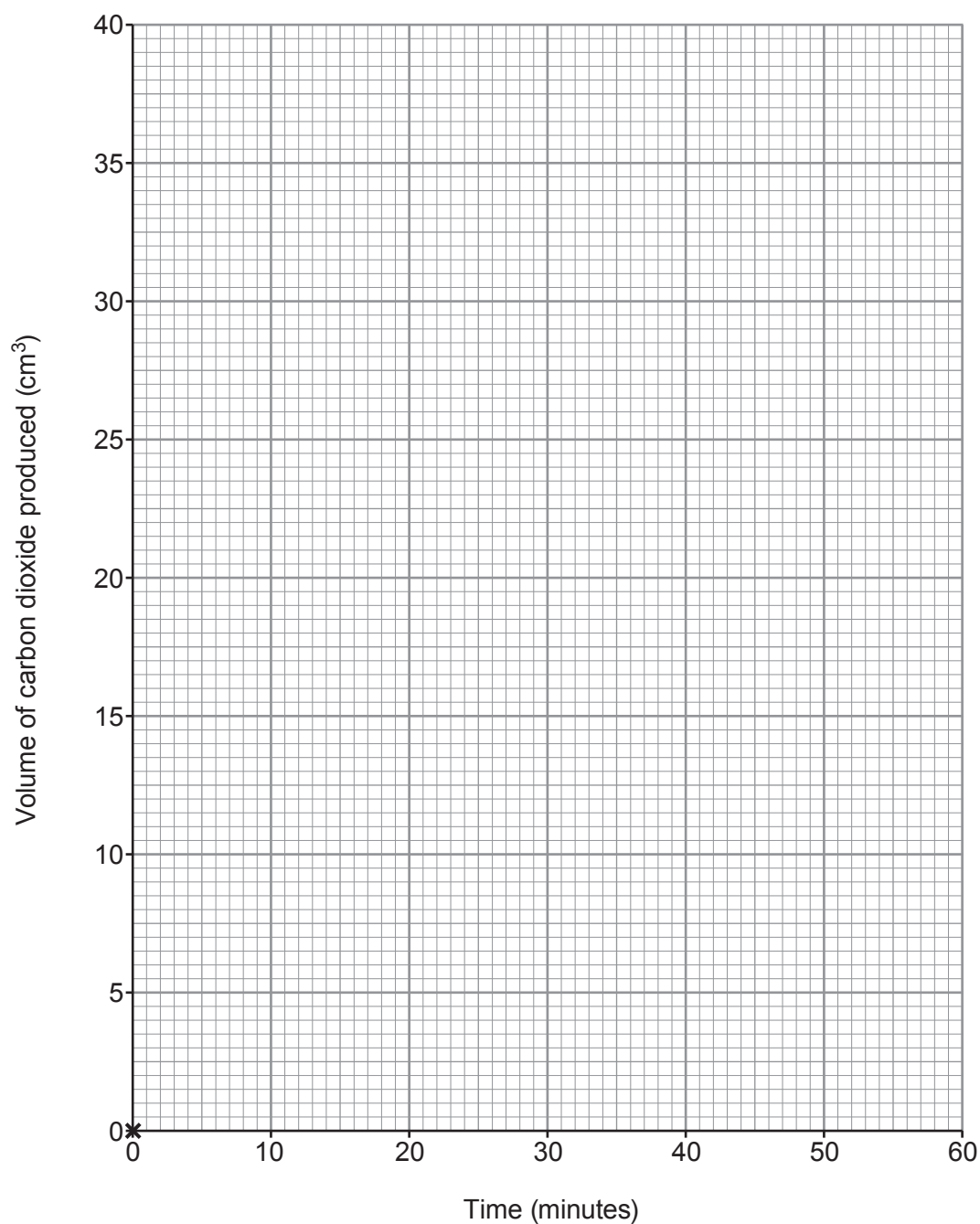
Time (minutes)	Volume of carbon dioxide produced (cm ³)
0	0
10	6
20	12
30	17
40	25
50	30
60	36



(i) Plot the data on the grid below and draw a suitable line.

The first point has been plotted for you.

[3]



- (ii) Use the graph to find the volume of carbon dioxide produced after 25 minutes. [1]

Volume cm^3

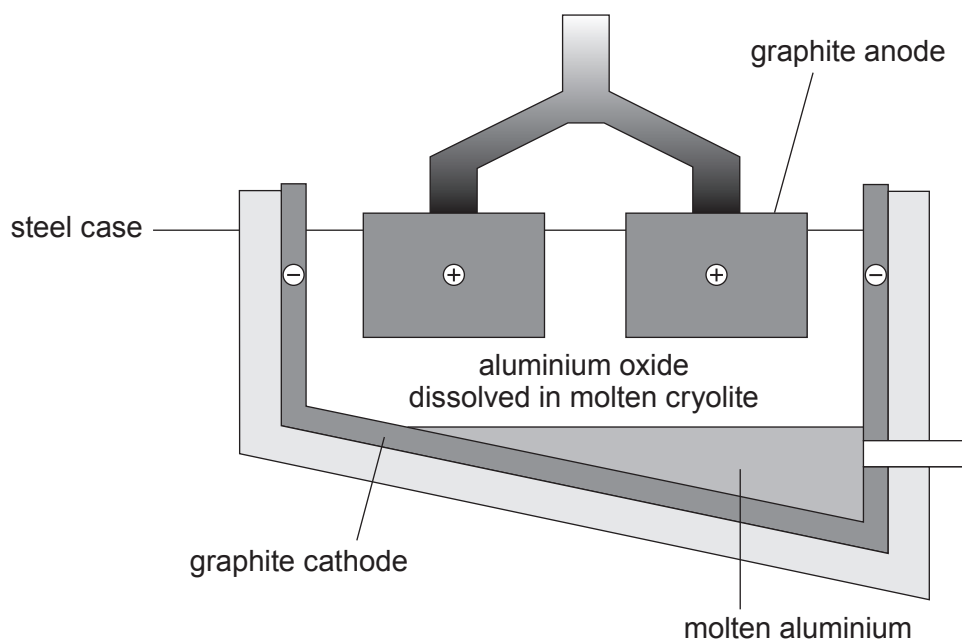
- (iii) Use the data to **estimate** how long it would take to produce 100 cm^3 of carbon dioxide gas. Assume that the rate of the reaction does not change. [1]

Time = minutes

9



4. (a) Aluminium metal is produced from aluminium oxide in a process called electrolysis. The diagram shows a cell used in electrolysis.



- (i) Underline the correct word in the brackets to complete each sentence. [4]

Aluminium oxide is a (**compound** / mixture / element) of aluminium and oxygen.

Cryolite is added to (**raise** / lower / maintain) the temperature used for electrolysis.

Electrolysis uses a lot of (**electrical** / solar / chemical) energy.

Aluminium is produced as a (**solid** / liquid / gas).



- (ii) Aluminium oxide contains the ions Al^{3+} and O^{2-} .

Give the formula of aluminium oxide.

[1]

.....

- (iii) Iron oxide is reduced by heating with carbon in a blast furnace.

Tick (✓) the box that states why aluminium oxide cannot be reduced in this way.

[1]

carbon is more reactive than aluminium	
iron is more reactive than aluminium	
aluminium is more reactive than carbon	

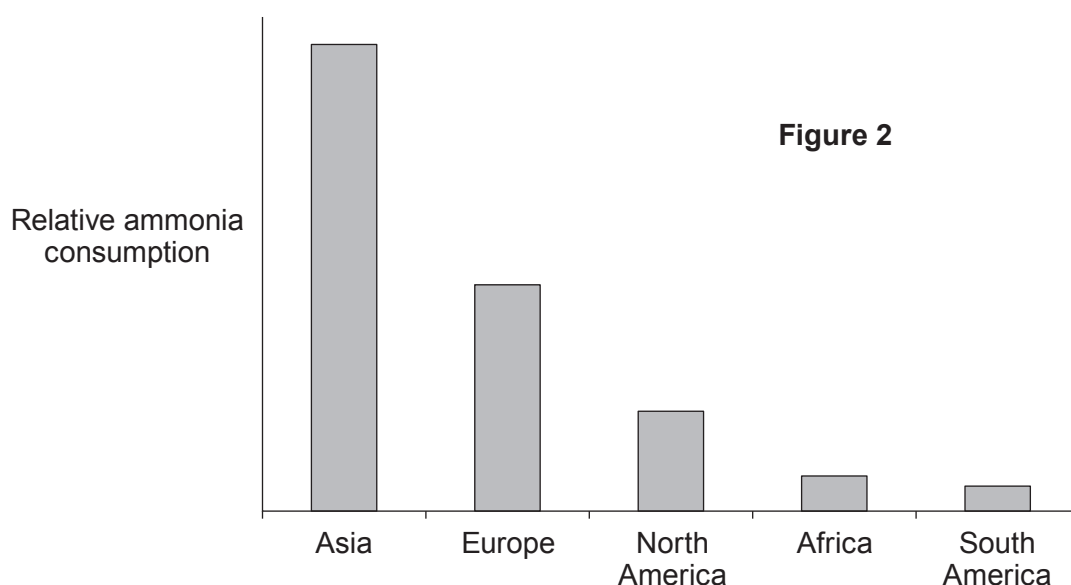
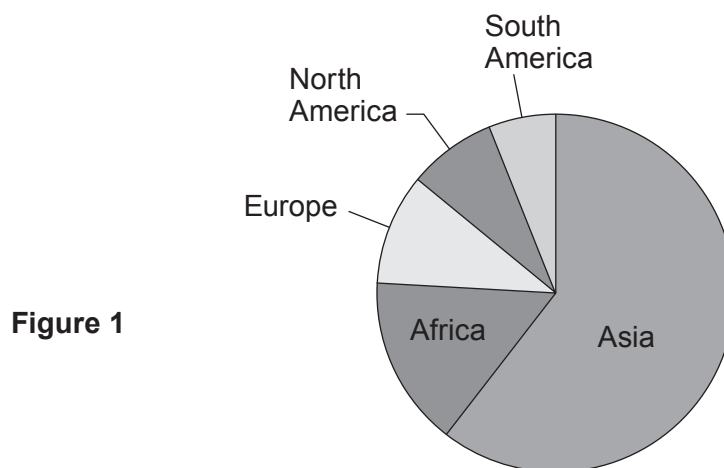


5. Plant and crop growth can be improved using fertilisers containing nitrogen. Fertilisers are substances that are added to the soil in order to increase the supply of nutrients that boost the growth of plants. With the rapid increase in global population, the demand for food has been rising tremendously. It is estimated that 40-60 % of agricultural crops are now grown with the use of different types of fertilisers. Many fertilisers are produced using ammonia.

When you use too much fertiliser in the soil, it can lead to eutrophication. Fertilisers contain substances like nitrates and phosphates that are washed into lakes, oceans and rivers by rain water. These substances lead to excessive growth of algae and plants in the waterways resulting in a decrease in the levels of oxygen. The decrease in oxygen levels leads to the death of fish and other aquatic animals and contributes to changes in food chains.

Figure 1 shows the relative populations of world continents in 2016.

Figure 2 shows the relative world consumption of ammonia in 2016.



- (a) (i) Tick (✓) **three** boxes that state the effects of fertiliser use which contribute to the death of aquatic animals in eutrophication. [2]

crop growth on fields increases	
fertilisers run into waterways	
plant growth in rivers and lakes increases	
aquatic animals do not have enough oxygen	
farmers' profits increase	

- (ii) State whether the information in **Figure 1** and **Figure 2** shows a link between the number of people living on each continent and the amount of ammonia used. Give reasons for your answer. [2]

.....

.....

.....

.....



- (b) The label on a bag of fertiliser shows the formula of the nitrogen compound that the fertiliser contains.



1.5 kg fertiliser

- (i) Give the name of the compound (NH₄)₂SO₄. [1]

.....

- (ii) Tick (✓) the box that gives the correct reason why plants need nitrogen. [1]

plants use nitrogen to make sugar	
plants use nitrogen to make water	
plants use nitrogen to make oxygen	
plants use nitrogen to make protein	

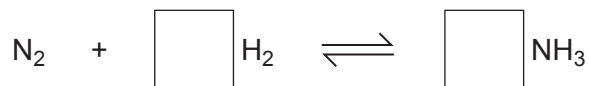
- (iii) 1.5 kg of fertiliser treats an area of 75 m². Calculate the mass needed to treat a lawn with an area of 15 m². Give your answer in **grams**. [2]

Mass = g



- (c) Ammonia is used to make many fertilisers. Ammonia gas is produced from nitrogen and hydrogen using the Haber process.

- (i) Balance the equation for the reaction between nitrogen and hydrogen in the Haber process. [1]



- (ii) State what is meant by the symbol $\rightleftharpoons$ used in the reaction equation. [1]

.....

- (iii) A catalyst is used in the Haber process.

- I. Give the name of the catalyst used. [1]

.....

- II. State why a catalyst is used. [1]

.....



6. (a) The table shows some information about substances **A-E**.

Substance	Soluble in water?	Malleable?	Boiling point (°C)	Conducts electricity?
A	no	no	4 200	no
B	yes	no	–79	no
C	yes	no	1 413	only when molten or in solution
D	no	yes	5 555	yes
E	no	yes	2 562	yes

(i) Give the **letters** of the **two** substances which are metals.

[1]

..... and

(ii) Give the **letter** of the substance which could be carbon dioxide.

[1]

.....

(iii) Give the **letter** of the substance which has a giant ionic structure.

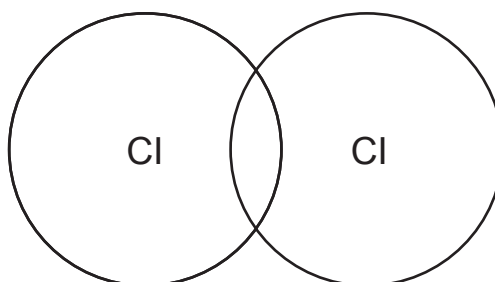
[1]

.....

(b) Complete the dot and cross diagram to show the outer electrons in a chlorine molecule, Cl₂.

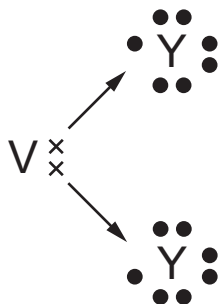
Each atom of chlorine has 7 electrons in its outer shell.

[2]



- (c) The diagram below shows the electronic changes that occur when atoms **V** and **Y** form ionic bonds. The dots and crosses show outer shell electrons.

V and **Y** are **not** the chemical symbols of the elements.



- (i) Give the **charge** on each ion produced. [2]

V

Y

- (ii) Circle the formula of the compound formed. [1]



- (iii) Potassium oxide, K_2O , is another ionic compound.

The relative formula mass of potassium oxide is 94.

Calculate the percentage of potassium in potassium oxide, K_2O . Give your answer to the nearest **whole number**. [2]

$$A_r(\text{K}) = 39$$

Percentage = %



7. (a) A student made some copper(II) sulfate crystals by reacting copper(II) carbonate powder with sulfuric acid using the following method.

Stage 1 Measure 50 cm³ of sulfuric acid into a beaker.

Stage 2 Add copper(II) carbonate powder, one spatula at a time, until all the acid has reacted.

Stage 3 Filter the mixture.

Stage 4 Obtain crystals from the solution.

- (i) State how you would carry out Stage 4 to get the largest possible crystals. [1]

.....
.....

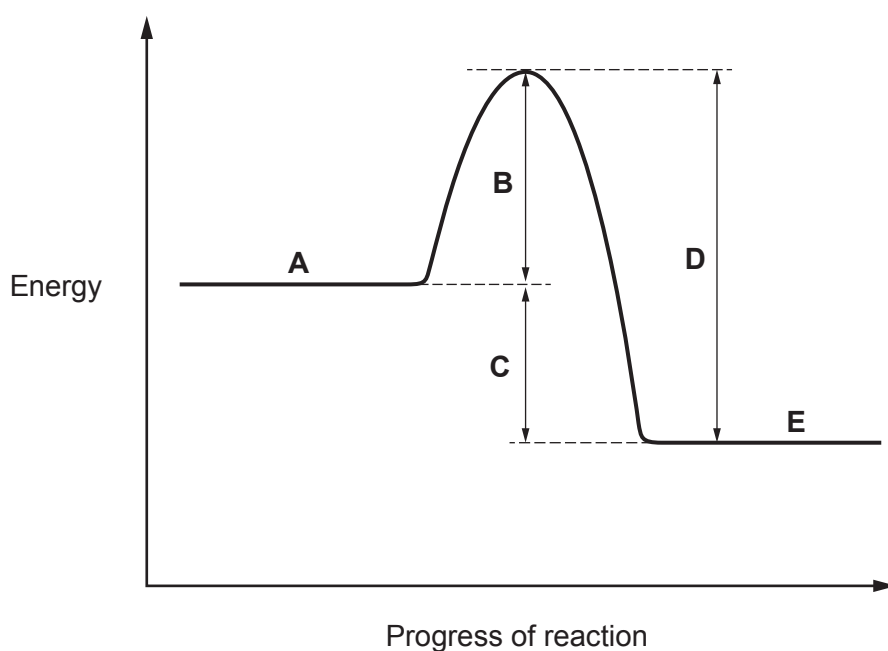
- (ii) Crystals of copper(II) sulfate could also be made using copper(II) oxide powder instead of copper(II) carbonate powder. State and explain how the observations in Stage 2 would be different. [2]

.....
.....
.....

- (iii) Complete the symbol equation for the reaction between copper(II) oxide and sulfuric acid. Copper(II) sulfate is one of the products. [2]



(b) The diagram shows an energy profile for a reaction.



(i) Give the **letter** that represents each of the following parts of the energy profile. [2]

Part of the energy profile	Letter
energy change for the reaction	
energy of the reactants	
activation energy of the reaction	

(ii) Give the meaning of the term activation energy. [1]

.....

.....

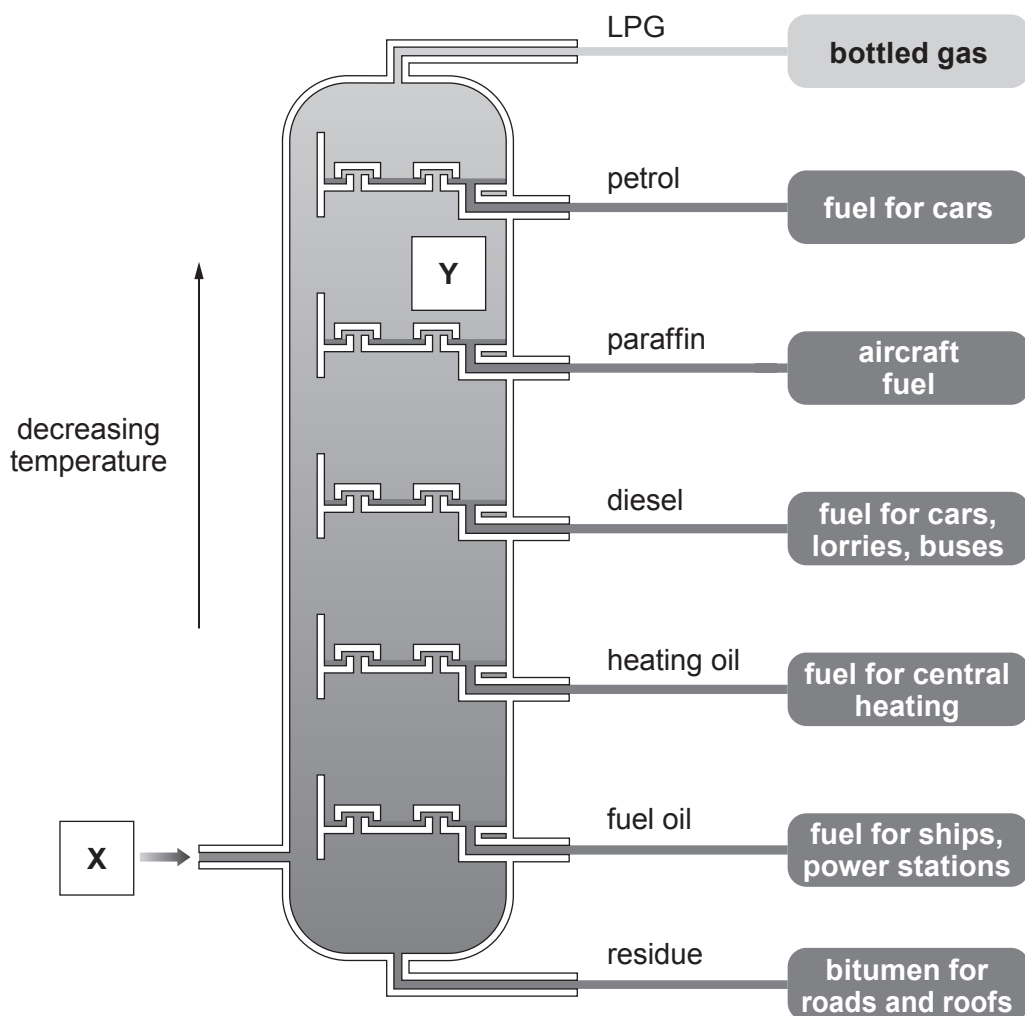
(iii) State how the energy profile shows that this is an exothermic reaction. [1]

.....

.....



8. (a) Crude oil is separated into mixtures of hydrocarbon compounds in the process of fractional distillation. Many of these fractions are used as fuels.



- (i) Name the changes of state happening at **X** and at **Y**.

[1]

X

Y

- (ii) Explain why different fractions are formed at different levels.

[2]

.....

.....

.....

.....



- (iii) A hydrocarbon fuel was burned and used to heat 100g of water. The water temperature rose from 18.5 °C to 38.2 °C.

Use the equation below to calculate the amount of energy released by this fuel. Give your answer to **two** significant figures. [3]

$$\text{energy (J)} = \text{mass of water (g)} \times 4.2 \times \text{temperature rise (}^{\circ}\text{C)}$$

Energy = J

- (b) The products of fractional distillation can undergo a process called cracking to produce smaller, more useful hydrocarbons.

- (i) Complete the equation for the cracking of $\text{C}_{16}\text{H}_{34}$. [1]



- (ii) State the **two** conditions used for cracking. [1]

.....

- (iii) The molecule with the formula C_2H_4 is an unsaturated hydrocarbon.

Give the meaning of the term unsaturated. [1]

.....

.....

- (iv) State why there is a high demand for each of the following products of the cracking reaction. [2]

octane / C_8H_{18}

ethene / C_2H_4

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



Group

0

1H Hydrogen 1										4He Helium 2																									
7Li Lithium 3		9Be Beryllium 4		11Na Sodium 11		12Mg Magnesium 12		13Al Aluminium 13		14Si Silicon 14		15P Phosphorus 15		16S Sulfur 16		17Cl Chlorine 17		18Ar Argon 18																	
19K Potassium 19		20Ca Calcium 20		21Sc Scandium 21		22Ti Titanium 22		23V Vanadium 23		24Cr Chromium 24		25Mn Manganese 25		26Fe Iron 26		27Co Cobalt 27		28Ni Nickel 28		29Cu Copper 29		30Zn Zinc 30		31Ga Gallium 31		32Ge Germanium 32		33As Arsenic 33		34Se Selenium 34		35Br Bromine 35		36Kr Krypton 36	
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87Fr Francium 87		88Ra Radium 88		89Ac Actinium 89																															
																			Key																

relative atomic mass

atomic number